Conversation, Not Delegation

How to Think With AI, Not Just Use It

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AI disclosure: This book was written using the methodology it describes. AI tools were used as thinking partners throughout the drafting, iterating, and refining process. The author reviewed, challenged, and took responsibility for every sentence. Full details of the tools used are in the front matter.

Companion website:
https://michael-borck.github.io/conversation-not-delegation
Source:
https://github.com/michael-borck/conversation-not-delegation

Table of contents

Preface 1

I Part 1: Understanding the Landscape 13

1 The Delegation Trap 15

2 What Is AI? 21

3 What Are Large Language Models? 27

4 Does AI Make Us Dumber? 37

II Part 2: Principles 45

5 AI Last 47

6 The Conversation Loop 53

7 Staying Critical 63

8 VET Your AI: The Push-Back Framework 73

III Part 3: The Methodology 81

9 Is Prompting Dead? 83

10 Prompt Chaining: Building on What You Started . . . 93

11 Eight Techniques for Deeper Thinking 105

12 Using AI to Help You Use AI 119

IV Part 4: Putting It Together 127

13 Conversations Across Disciplines 129

14 When the Loop Runs Itself (Advanced and Optional) . 137

15 Becoming More Capable 143

Appendices **149**
A Prompt Structuring Frameworks 149
B Quick Reference Cards 155
C Further Reading 165
About the Author 169

Preface

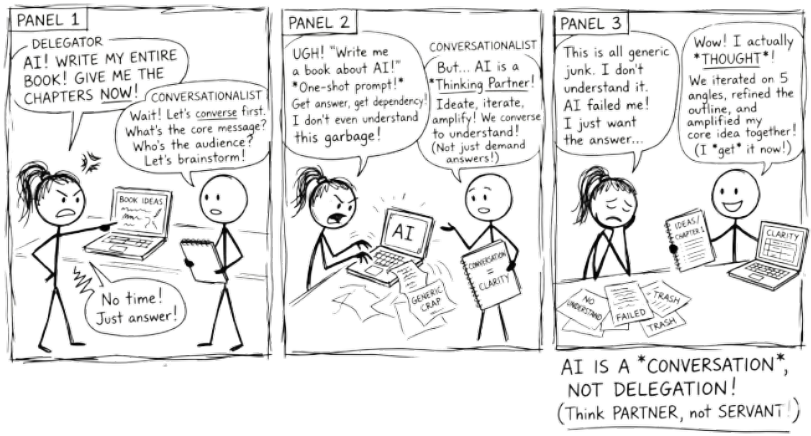


Figure 1: Delegation is quick. Conversation is how you actually learn.

Why This Book Exists

I wrote this book because I kept watching smart people use AI badly.

Not badly in a technical sense. They could find the tools, type the prompts, and get outputs. But they were delegating their thinking to a machine and accepting whatever came back. The AI was doing the work, and the human was doing the clicking. The outputs looked polished. The thinking behind them was hollow.

There is a different way. Instead of delegating to AI, you can converse with it. You can use it to brainstorm, to challenge your assumptions, to stress-test your reasoning, to explore angles you would not have

considered alone. When you do this, the AI does not replace your thinking. It amplifies it. This book teaches you how.

The academic literature has other names for what gets preserved when you do this. Philosophers and ethicists call it **agency**: your capacity to choose, weigh, and own the outcome. AI policy and system design call it **human-in-the-loop** (HIL): the principle that a person remains an active participant in decisions a system could otherwise automate. This book uses a different word, **conversation**, because the academic terms describe *what* you keep, and the conversational frame describes *how* you keep it. Same idea, three vocabularies. If you came in looking for guidance on human agency or HIL, you are in the right place. See Appendix B for the terms map.

Who This Book Is For

This book is for anyone who uses AI as a thinking tool. Students, professionals, researchers, educators: if you interact with large language models as part of your work or study, this book will change how you approach those interactions.

You do not need a technical background. You do not need to write code. You need curiosity, a willingness to think critically, and a problem worth solving.

If you work with AI in any capacity, or plan to, this book will give you a framework for doing it well.

What This Book Is Not

This is not a guide to specific AI tools. It does not teach you how to use ChatGPT, Claude, or any particular platform. Interfaces change; the principles here do not. Where the book names a specific tool or describes how current models behave, those details will date faster than the surrounding ideas. Keep the principles and expect the interfaces to change.

It is not a prompt engineering manual. You will learn to structure prompts well, but the book argues that the prompt begins the conversation; it does not replace it. If you are looking for a library of copy-paste prompts to get finished outputs in one shot, this is the wrong book.

A related warning: conversation, in this book, is not a string of zero-shot prompts run back-to-back. That is just delegation in volume. Real conversation requires pushback, course correction, and your judgement woven through each exchange. Chapter 6 draws the distinction carefully.

It is not about AI apps that embed intelligence into other software, such as writing assistants, email autocompletes, and coding copilots. Those tools run on prompts someone else wrote. This book is about conversations you control yourself. Both have their place, but they are different activities.

This book argues the opposite: you will solve your problems, and AI can help you think more clearly along the way.

If You Are Feeling Uncertain

Many people experience a quiet anxiety: a sense that everyone else has figured this out already, that the technology is moving too fast, that they are falling behind. That feeling is nearly universal and rarely admitted. You are not behind. The technology is genuinely new, it is genuinely confusing, and the people who appear to have it all figured out are mostly just a few weeks ahead. This book meets you wherever you are.

How This Book Is Structured

Part 1 begins with the delegation trap, then gives you enough understanding of AI and large language models to recognise why that trap is so tempting. Part 2 establishes the principles that make AI conversations productive rather than passive: think first, work in a conversation loop, stay critical, and verify what matters. Part 3 turns

those principles into practical prompting methods and techniques you can use immediately. Part 4 shows the approach working across disciplines, offers an optional advanced chapter on autonomous loops, and closes with the argument that matters most: the goal is not to get AI to do more, but to become more capable yourself.

By the end, you should be able to:

- decide whether a task calls for mechanical delegation, conversation, a more reliable non-AI tool, or no AI at all;
- frame a useful starting request without mistaking prompt structure for judgement;
- identify and redirect a plausible but unsuitable response;
- verify the claims that carry real consequences; and
- produce a record showing which parts came from the tool, which decisions were yours, and why you can defend the result.

Evidence of progress, part by part

Part	What you should be able to show
Understanding the landscape	Classify a task, distinguish a model from its surrounding tools, and explain why a plausible answer needs checking
Principles	Preserve a first attempt, identify a consequential error, and complete Verify–Explain–Test against independent evidence
Methodology	Write a scoped brief, choose a technique for a specific difficulty, and place a check between dependent steps

Part	What you should be able to show
Putting it together	Produce an owned decision with source support, a rejected or revised suggestion, and a boundary test

Keep a small learning record as you read. Start with one task and your unaided attempt; add the evidence or correction each part makes possible. The worked community workshop in Chapter 13 shows the complete record and supplies a parallel task with an answer check.

You can complete the independent activities without an AI account. "Unaided" here means without handing the judgement to AI; calculators, assistive technology, notes, dictation, and help accessing the material are welcome. Shorten a session or explain with a diagram if that lets you inspect the reasoning more clearly. The advanced loop chapter is optional, not a gate to completing the book.

How to Read This Book

This is not a textbook and it is not a research paper. The ideas here reflect one practitioner's perspective, informed by research, experience, and ongoing experimentation. They are starting points, baselines to work from as you discover your own approach to AI.

Where claims rest on published research, you will find pointers in the Further Reading appendix (Appendix C). Where they rest on practice and observation, that should be obvious from the writing. In both cases, the advice in this book applies to itself: engage with it critically, push back where your experience says otherwise, and make the ideas your own. A book about not delegating your thinking to AI would be contradicting itself if it asked you to accept its arguments uncritically.

Conventions Used in This Book

Throughout the book you will encounter coloured callout boxes. Each serves a different purpose.

> **Try this**
>
> **Green boxes** are hands-on exercises you can do right now, usually in two to ten minutes. They give you a way to experience the concept rather than just read about it.

> **Key idea**
>
> **Blue boxes** highlight important observations, ready-to-use prompts, or ideas worth pausing on.

> **Watch out**
>
> **Yellow boxes** flag common mistakes, weak examples, or things that look right but are not.

> **Critical point**
>
> **Red boxes** mark habits or principles that are essential. Do not skip these.

You will also find tables throughout the book that compare approaches, map strategies across domains, or summarise frameworks. These are reference material, worth bookmarking, not just reading once.

Most chapters close with a short practice section. Selected chapters provide complete RTCF starting prompts; other chapters ask you to begin without AI, evaluate evidence, or transfer the method to your own work. RTCF (Role, Task, Context, Format) is covered properly in Chapter 9 and summarised in Appendix B. The structure is scaffolding, not a ritual. Use only the elements that clarify your task, then push back on the first answer.

How This Book Was Written

This book was written using the methodology it describes.

It began as a set of Python notebooks on computational thinking, practical exercises developed for teaching. Those notebooks evolved, through multiple conversation loops with AI, into the sibling book *Converse Python, Partner AI*. That process surfaced a methodology that was broader than programming: the principles of conversation over delegation, of staying critical, of using AI to amplify thinking rather than replace it. This book is the result of extracting and generalising that methodology for a wider audience.

At every stage, the process looked like what you will read about in Chapter 6. Ideas were brainstormed, challenged, restructured, and refined across dozens of conversations. AI drafted, the author pushed back. AI suggested structures, the author broke them and rebuilt them. The frameworks in this book (RTCF, VET, the eight techniques) were tested in the writing of the book itself. Where AI was sycophantic, it was told to stop. Where it was generic, it was pressed for specifics. Where it was wrong, it was corrected.

You may be thinking: if this book was written with AI, why should I trust it? That reaction is understandable, and this book has a name for it: the AI Dismissal Fallacy (Chapter 7). The quality of an idea does not depend on whether a human or a machine contributed to its development. What matters is whether the author can explain, defend, and stand behind every sentence. The answer is yes, because conversation, real conversation with pushback and judgement, is how the thinking was done.

Transparency about process is more important than the comfort of pretending AI was not involved. Every professional will face this question soon enough. This book's answer is to show the work.

Tools used in producing this book:

- **Claude** (Anthropic): conversational AI partner for drafting, iterating, and refining text
- **Claude Code**: command-line tool for managing files, building, and publishing

- **Google Gemini (Nano Banana 2)**: generative AI for creating the comic strip illustrations (Chapters 1–8)
- **FLUX Playground** (Black Forest Labs): generative AI for creating the comic strip illustrations (Chapters 9–14)
- **Python** and **Jupyter Notebooks**: the original medium in which the ideas were developed and tested; also used to generate the cover image (`create_cover.py`)
- **Quarto**: open-source publishing system used to produce the HTML, PDF, and epub editions
- **Git and GitHub**: version control and hosting, including GitHub Pages for the web edition

Ways to Engage with This Book

A book about conversation should be available in more than one form. Pick the format that fits how you think and learn.

- **Read it online.** The full book is freely available at the companion website, with dark mode, search, and navigation.
- **Read it on paper or e-reader.** Available as a paperback and ebook through Amazon KDP, for those who prefer to read offline or away from a screen.
- **Download the PDF or epub.** Generated from the same source, available from the website.
- **Converse with it.** The online edition includes a chatbot grounded in the book's content. Ask it questions, challenge its answers, and practise the methodology on the methodology itself. You can also use the `llm.txt` file to paste the entire book into ChatGPT, Claude, or any AI tool for a deeper conversation.
- **Listen to it.** Upload the `llm.txt` file to Google NotebookLM and generate an audio overview or podcast-style discussion of the content.
- **Explore the source.** The full source is on GitHub (https://github.com/michael-borck/conversation-not-delegation), including every chapter, the build system, and the revision history. DeepWiki (https://deepwiki.com/michael-

borck/conversation-not-delegation) provides an AI-navigable view of the repository.
- **Browse all books.** This book is part of a series. See all titles at books.borck.education (https://books.borck.education).

The online version is always the most current. The printed and ebook editions are updated periodically. Use the free online edition and its downloads for the latest available revision. A purchased ebook is not guaranteed to update automatically.

The Series

The series began as a set of conversations and domain books. The trunk became visible as those books were written: the same habits kept returning under different names. Naming the trunk now makes the relationship explicit. It does not make the series a hierarchy.

Conversation, Not Delegation is the method. *The Human Edge* (https://michael-borck.github.io/the-human-edge-using/) is the practice, applying that judgement to everyday professional work. The domain books are branches: each takes the same habits into a particular craft. The order below is one reading path.

Route	Start here	Continue with
Concepts before code	Conversation, Not Delegation	Converse Python, Partner AI; apply its method alongside the appropriate programming level
Python from the beginning	Think Python, Direct AI	Code Python, Consult AI, then Ship Python, Orchestrate AI
Web development	Think Python, Direct AI	Build Web, Guide AI

Two ways in: start with *Conversation, Not Delegation* if you want the

ideas before (or without) code, or start with *Think Python, Direct AI* if you want to code from your first page. The methodology in *Converse Python, Partner AI* applies to every path.

- **Conversation, Not Delegation**: the general methodology for working with AI across any discipline. No code required; the broadest entry point.
- **Think Python, Direct AI**: computational thinking for absolute beginners. The entry point if you want to code from the start.
- **Converse Python, Partner AI**: the intentional prompting methodology applied to software development. A companion you can read alongside any Python book.
- **Code Python, Consult AI**: focused Python fundamentals through AI conversation.
- **Ship Python, Orchestrate AI**: professional Python development practices and tooling.
- **The Absolute Minimum You Must Know**: the free, always-current web companion at minimum.borck.education (https://minimum.borck.education): quick primers on every topic this series covers in depth.
- **Build Web, Guide AI**: web development with AI as your development partner.
- **The Human Edge**: the workplace companion, applying the same judgement to everyday professional work, briefing, checking, and delivery.
- **Speak Software**: programming at the natural-language layer, for readers who want to direct agents and build with them.

All titles are available at books.borck.education (https://books.borck.education).

You are reading *Conversation, Not Delegation*, the methodological trunk of the series.

In the Python track, this book has a companion: *Converse Python, Partner AI*, which covers the same methodology through the lens of Python programming. If you are a developer or data scientist, that book may be a better fit. If you are not, you are in the right place. Both books are independently complete. You do not need one to

benefit from the other.

Part I

Part 1: Understanding the Landscape

Chapter 1

The Delegation Trap

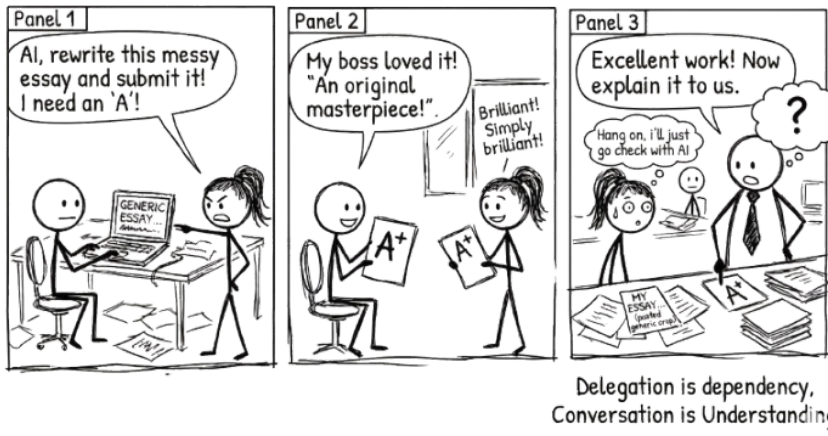


Figure 1.1: Delegation is dependency, Conversation is Understanding.

> A single prompt gives you a single answer. A **conversation** gives you **understanding**.

1.1 The result you cannot explain

Imagine taking a finished proposal into a meeting. The writing is clear, the recommendation is confident, and it took you minutes to

produce. Then someone asks why you ruled out the cheaper option. You cannot answer without reopening the chat.

The problem is not that a tool helped write the proposal. It is that you accepted a decision without acquiring the understanding needed to defend it. This book calls that the **delegation trap**.

A fluent conversation can leave you in the same position. Ten follow-up prompts are no guarantee of learning. The useful distinction is what you can check, explain, and decide after the interaction.

1.2 Two ways to ask for help

Compare these requests. Neither column guarantees a particular outcome; the second makes your reasoning more visible.

Task	Output request	Request that invites scrutiny
Marketing plan	"Write a launch plan."	"Here is my proposed audience and budget. Which assumption most threatens the plan?"
Reading a paper	"Summarise this paper."	"Compare my summary with the paper. Which claim did I overstate?"
Community event	"Plan the workshop."	"Check my timetable against room capacity, reset time, and budget. Mark anything you cannot establish."

An output request may be entirely appropriate. A summary can make a difficult document accessible or help you decide whether to read it. A generated first draft can help someone who is unsure

how to start. The question is whether the chosen help serves your purpose and leaves important checks with someone able to perform them.

If your purpose is to learn, preserve opportunities to attempt, explain, and test. If your purpose is simply to reformat an already checked table, learning a new skill may not be the goal.

1.3 Responsibility stays somewhere

A model can propose a choice, score alternatives, or execute an authorised action. That does not settle who is accountable for the result. Decide who owns the decision, what they must inspect, and when another person's expertise is needed.

In the proposal example, you might ask the tool to list the alternatives it omitted. That list is a starting point for checking, not proof that it found them all. You still need the source prices, the client's constraints, and a defensible comparison.

Avoid treating a request for "reasoning" as access to the machine's private workings. An explanation is another output. Check whether its claims and calculations support the recommendation.

The practical standard is modest: can the responsible person give a defensible account of the decision using evidence outside the model's assertion?

1.4 Productivity and learning are different outcomes

A finished deliverable matters. So do time, accessibility, quality, and the capability you need for later work. You do not have to pretend that output is worthless to care about learning.

Keep these outcomes separate when comparing workflows:

- Did the task get completed to the required standard?

- How much time did drafting, checking, and correction actually take?
- Can you explain the important decisions?
- Can you apply the method to a new case with the support appropriate to that task?

A fast draft followed by slow repair may not save time. A slower practice session may be worthwhile when you are learning. A trusted accessibility tool may remain useful indefinitely. None of those observations makes independence from every tool a moral test.

Research does raise concerns about some kinds of reliance. It does not establish that every delegated task weakens a mental muscle or that conversation invariably improves skill. Chapter 4 separates assisted performance from subsequent learning and discusses the evidence and its limits.

1.5 Decide what to hand over

Start with the intended use and consequences. For checkable mechanical work, delegation may be sensible: use a calculator for arithmetic, or a deterministic conversion tool for a file format when available. Check the result in proportion to the risk.

For unfamiliar or consequential work, identify the missing knowledge before accepting an answer. Conversation can help you frame questions for a qualified person; it cannot manufacture that person's approval. Some tasks should stay out of an unapproved AI service entirely.

The decision map in Appendix B brings these choices together. For now, name one check that would catch a plausible failure in your current task.

> Try this (2 minutes)
>
> Take a recent output and mark one claim you could independently verify. Find its supporting source or calculation. If you cannot, record the gap rather than asking the model to

reassure you.

1.6 Begin a learning record

Choose an ordinary task you will carry through the book. Write its purpose, what you already think, and what would count as a useful result. Keep the first attempt; it gives you something concrete to compare with the final work.

You can use the community workshop in Chapter 13 if you do not have a suitable task. Its printed constraints and deliberately flawed draft let you practise without an AI account.

The aim is an owned result: useful work whose important claims you can check and whose decisions you can explain. The following chapters give you ways to build that habit.

1.7 Take it to the AI

Where to start. Three questions worth arguing about rather than looking up:

1. Take a task you handed to AI this week and ask what you would have learned by doing it yourself. Then decide whether the trade was worth it. Sometimes it was.
2. Ask it to rewrite three of your delegation-shaped prompts as conversation-shaped ones. The pattern usually becomes visible by the second.
3. Ask where the line falls between mechanical delegation, which is fine, and delegating judgement, which is the trap, using examples from your actual job rather than generic ones.

Keywords to steer with. delegation versus conversation mindset · judgement · agency · process over product · output volume · cognitive offload · the delegation test · mechanical versus expert tasks

A starting prompt.

Role: You are a learning adviser helping me choose appropriate support and checks.

Task: Sort the tasks I list into three piles: safe to delegate, worth conversing about, and the ones I should be doing entirely alone.

Context: I have read a chapter arguing that mechanical delegation is fine and delegating judgement is the trap, and that the test is whether a task requires my expertise. Here is what I handed to AI in the last week: [list it]. My role is [describe it].

Format: Three lists, one line of reasoning against each item. Then identify one task worth practising independently and one where delegation may have been appropriate. Mark assumptions instead of diagnosing skill loss.

Chapter 2

What Is AI?

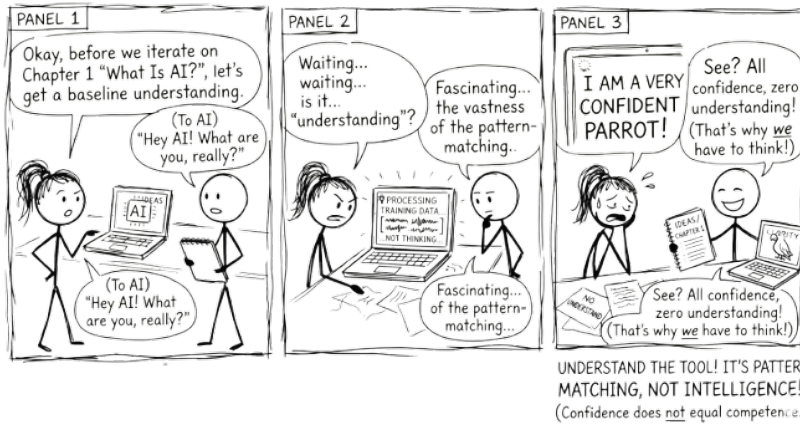


Figure 2.1: Understand the tool! It's pattern matching, not intelligence.

> The gap between what AI appears to do and what it actually does is where good decisions begin.

You might be wondering why this book pauses after the opening problem to explain what AI is. The reason is practical: every framework in this book, every technique for prompting, verifying, and staying critical, exists because of something specific about how these tools work. If you understand the mechanics, even at a high level,

the techniques stop feeling like arbitrary rules and start feeling obvious. You will know *why* you need to verify AI output, not just that you should.

Artificial intelligence is software that can perform tasks that typically require human intelligence. There is no magic or consciousness in that, just software that can do things that used to require a human brain.

When we say "intelligence" here, we mean specific capabilities: recognising patterns in images, making predictions from data, understanding language, generating text. AI does not "think" the way you do. It processes data using mathematical patterns. But the results can look remarkably intelligent, and that gap between appearance and reality is worth understanding from the start.

2.1 Why Now?

AI is not new. The 1956 Dartmouth workshop was an early milestone. Researchers have been working on it for decades. So why is it suddenly everywhere?

Three things converged.

2.1.1 We have massive amounts of data

Every click, purchase, photo, and search creates data. Organisations now have billions of examples to learn from. Netflix has billions of viewing decisions. Wikipedia has millions of articles. Large collections have enabled new training approaches, but access, quality, permissions, and representation still constrain useful data.

2.1.2 Computing power got cheap and fast

Specialised hardware and shared computing infrastructure have made many kinds of training and inference more practical. Cost still depends on the model and workload; large-scale training is not equally accessible to everyone.

2.1.3 Better algorithms were discovered

New mathematical techniques, particularly a family called "deep learning," dramatically improved what AI could do with data and compute. Researchers figured out how to stack layers of pattern recognition so that each layer builds on the one below it. The result was AI that could handle far more complex tasks than previous approaches allowed.

These three forces fed each other. More data made better algorithms worthwhile. Cheaper compute made bigger models practical. Better algorithms made use of both. The result is that AI which was science fiction ten years ago is now practical, affordable, and widely deployed.

2.2 The AI Family Tree

AI is not one thing. It is a family of technologies, each suited to different problems.

Rule-based systems are the oldest approach. Programmers write explicit rules: if this condition, then that action. These work well for clear, logical decisions that do not change much. Tax software and simple chatbots use this approach. The limitation is obvious: someone has to anticipate every scenario in advance.

Machine learning flips the process. Instead of writing rules, you show the system thousands of examples and let it figure out the patterns. Show it ten thousand photos labelled "cat" and ten thousand labelled "not cat," and it learns to tell the difference. This works well for pattern recognition, predictions, and problems where the rules are too complex to write by hand. The catch is that it needs lots of good examples and it is only as good as the data you give it.

Deep learning extends machine learning with multiple layers of pattern recognition, loosely inspired by how neurons work. The first layer might detect edges in an image. The second combines edges into shapes. The third combines shapes into objects. The fourth recognises that the object is a golden retriever. This layered approach

handles very complex tasks: understanding speech, translating languages, driving cars, and generating text.

Large language models are the branch this book focuses on. They are covered in depth in the next chapter.

These approaches coexist. Deep learning is a family within machine learning, and many language models use deep neural networks. A single application can combine explicit rules, learned models, and conventional software.

2.3 What AI Can and Cannot Do

AI is strong where humans are slow and slow where humans are fast.

AI handles well:

- Repetitive pattern recognition across large volumes of data
- Processing and summarising massive amounts of information
- Tasks with clear success metrics and well-defined inputs
- Narrow, specific problems: classify this, translate that, predict the next value

AI struggles with:

- Inputs or conditions poorly represented in an evaluation
- Claims needing current, authoritative evidence that the system lacks
- Explanations whose faithfulness to the underlying computation is unknown
- Decisions needing contextual knowledge, value choices, or human authority

These are evaluation questions, not fixed claims that machines can never reason, generalise, or create something novel.

AI is not a replacement for human thinking. It is a tool that is extraordinarily capable in some domains and brittle in others. The people who use it well are the ones who understand where that line falls.

2.4 Hype Versus Reality

You will hear that AI will replace all jobs, that it is smarter than humans, that it can solve any problem. None of this is accurate.

AI can automate tasks within jobs, and this can also change or eliminate roles. New tasks may emerge, but the net outcome depends on the setting. The radiologist still reads the scan, but AI flags the areas worth examining. The writer still shapes the argument, but AI helps with research and drafts.

Some AI systems are specialised; others support many tasks and modalities. A system's name or performance on one task does not establish its suitability or permissions for another.

AI is a tool, not a strategy. Saying "we need an AI strategy" is like saying "we need a spreadsheet strategy." The question is always: what problem are you solving? AI is one way to solve it.

And AI reflects its training data. If that data contains biases, the AI learns those biases. It is not objective. It is a mirror of the information it was trained on, with all the distortions that implies.

> Key insight
>
> Automating tasks can also change jobs. AI is one tool among many. Start with the problem, not the technology.

2.5 The Practical Framing

Think of AI, as you read this book, as a capable but literal collaborator.

Capable, because it can process information, generate text, spot patterns, and respond to instructions at a speed and scale no human can match. Literal, because it does exactly what the patterns in its training suggest, not what you meant, not what would be wise, not what the situation actually calls for.

Clear instructions and useful context can help. Results also depend

on the model, application, data, and task. A structured conversation does not guarantee a correct or useful result.

This book is about thinking better with AI as a partner.

2.5.1 A Note on AI Apps and Hidden Prompts

You will increasingly encounter AI not as a blank conversation window but embedded inside other tools: a writing assistant in your word processor, an autocomplete in your email, a copilot in your spreadsheet. These tools are convenient, and many are genuinely useful.

Their instructions and available controls vary. Some expose editable prompts or context; others offer only a few settings. Inspect what you can supply, change, and verify instead of assuming every embedded tool prevents meaningful steering.

Choose an interface suited to the task and its data constraints. Direct chat is useful for open-ended dialogue; a constrained interface may be preferable for a well-defined workflow. In either case, check the result and understand what an acceptance button will do.

2.6 Independent practice: classify a task

Time: 15–20 minutes. Write three tasks: sorting a supplied list, drafting an announcement, and deciding whether an organisation may share personal records. For each, name the required capability, permission, and evidence. Do not infer how a product works from its branding.

Artefact: a three-row task map. **Success:** distinguish a possible output from permission to act; identify a non-AI or specialist route where appropriate. No AI account is needed to complete the independent version.

Chapter 3

What Are Large Language Models?

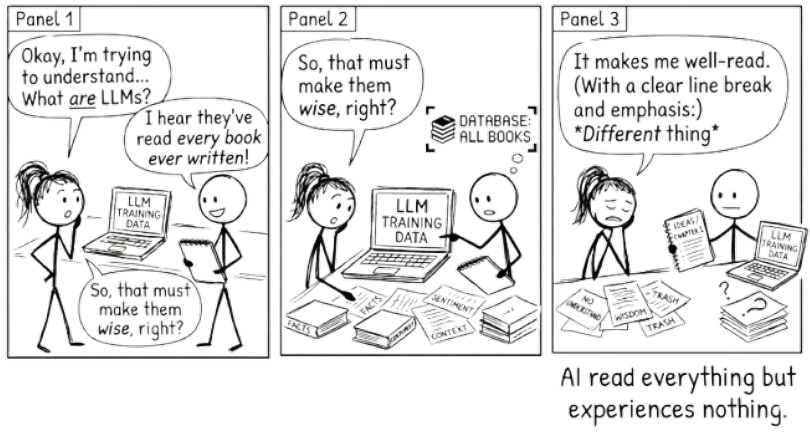


Figure 3.1: AI read everything but experiences nothing.

> Fluent language is useful. Its accuracy still needs evidence.

The opening comic exaggerates for effect: no model has read everything. Its training material and the information available in a particular conversation are limited.

Understanding how LLMs actually work is the foundation for every practical decision you will make when using them. Why do they hallucinate? Because they predict plausible text, not true text. Why does a structured prompt work better? Because it gives the model better patterns to predict from. Why do you need to verify output? Because an answer's fluency does not establish its accuracy. The frameworks later in this book (RTCF, VET, the two-chat workflow, the average-versus-precise grid) all follow directly from what you learn here.

A common language-model training task is predicting the next token, a piece of text. Conversational systems may also undergo further training for instruction following and other behaviours. Prediction is an important foundation, not a complete description of every model or application.

From that single task, predicting the next word, these models learned to write essays, answer questions, translate languages, summarise documents, explain complex topics, and hold conversations. The gap between "predict the next word" and "do all of that" is where the interesting story lives.

3.1 How Predicting Words Becomes Something More

Think about what it takes to predict the next word well.

"The cat sat on the ___." Easy. You guess "mat" because you have seen the pattern before.

"The capital of France is ___." To get this right, you need to have absorbed a fact about the world.

"If you drop a glass, it will ___." Now you need cause and effect.

"The patient presented with fever, joint pain, and a butterfly-shaped rash, which is consistent with ___." Now you need specialised knowledge and the ability to connect symptoms to diagnoses.

Each of these is still word prediction. But to predict well, the model

had to learn grammar, facts, reasoning patterns, argumentation structures, and different styles of communication. The training objective was simple, but what emerged from it was not.

LLMs were trained on enormous amounts of text: books, articles, websites, academic papers, forums, code repositories. Billions of documents. By learning to predict what comes next across all of that text, they built an internal model of how language works, and of much of what language describes.

3.2 What "large" leaves unspecified

Models have learned numerical parameters, not a separate labelled fact in each parameter. Training data, model architecture, computation, and later training all affect performance. A larger model is not automatically better for every task.

Do not infer a product's training corpus, cost, or parameter count from its fluent answers. These details may not be public, and models described as "large" vary considerably. For your task, a measured comparison with suitable inputs is more useful than a size label.

3.3 Context and generation are different

Many conversational models generate an answer one token at a time, using the supplied context and the answer generated so far. Tokens are pieces of text, not always whole words. In a causal decoder, an output position does not attend to future output tokens. That does **not** mean the model cannot refer back to the prompt, or that every language-model architecture processes information in exactly the same way. Vaswani et al., Attention Is All You Need (https://arxiv.org/abs/1706.03762)

The product also decides what context reaches the model: instructions, conversation history, source passages, tool results, or a summary. Repeating a question can sometimes change an answer, but it is not a universal repair for an architectural inability to reread. Test whether the response uses the information you supplied.

3.4 What They Can Do

LLMs are strong at tasks that involve language and pattern recognition.

They can generate text in nearly any style or format: emails, reports, stories, technical documentation. They can also summarise long documents, extract key points, and explain complex ideas at different levels of sophistication. Translation between languages goes beyond word-by-word substitution to meaning and context. And because code is a language with patterns too, they can write and debug it.

An explanation of photosynthesis may be generated from learned patterns, supplied sources, or retrieved material. You need information about the application to distinguish these routes. A readable explanation is useful, but its content still needs checking.

3.5 What They Cannot Do

They can hallucinate. Models may state false information confidently, with or without access to sources. They can also express uncertainty or refuse. Neither fluency nor a cautious tone establishes whether the answer is correct; check consequential claims.

They reflect the biases in their training data. If the text they learned from contains stereotypes, blind spots, or skewed perspectives, those patterns show up in their outputs. Further training may reduce some patterns without eliminating all of them. Do not assume a model is either free of bias or identical to every other model. See Chapter 7 for a practical way to surface them.

They have no access to real-time information. An LLM's knowledge stops at its training cutoff. Recent information must arrive through an update, supplied context, or an external source; a training cutoff alone does not describe everything available in a conversation.

These limits are fundamental to what these systems are: an LLM is a model of language, not a model of truth.

3.6 Generation is not verification

A language model generates a continuation from learned parameters and the context it receives. That is different from fetching an authoritative record. It can generate a correct fact, a useful new combination, or an invented detail with similar fluency.

A surrounding system may also retrieve documents, search the web, or call a database. Retrieval-augmented generation deliberately combines a generator with retrieved material. Check whether retrieval actually occurred and whether the selected passage supports the claim. A citation-shaped string is not proof of either. Lewis et al., 2020 (https://arxiv.org/abs/2005.11401)

Some models can reproduce memorised training text, so "it never retrieves or retains anything" is also an unsafe shortcut. Memorisation, external retrieval, and generation are different mechanisms. Carlini et al., 2021 (https://arxiv.org/abs/2012.07805)

"Interpolation" can be a rough metaphor for familiar patterns. It is not a proof that models can never generalise, generate novelty, or solve a previously unseen problem. The practical boundary is narrower: a plausible answer does not establish correctness, provenance, or fitness for your use. Those require checks.

3.7 Match the check to the intended use

A private brainstorm can tolerate suggestions you discard. An exact quotation or calculation needs a source or reproducible check. A long task may also need intermediate checks so that an early mistake does not propagate.

Complexity and consequences are separate. A short refund promise can have larger consequences than a long private outline. The companion *Prove, Don't Persuade* uses a precision-and-consequence grid to decide review intensity; it is not a model-performance law.

For this book, use the decision map in Appendix B. Ask what must be correct, how you can check it, and who is affected if it is wrong.

No brand or average benchmark score answers those questions for you.

3.8 The model is not the application

A chat application may add instructions, conversation history, retrieval, memory, filters, or action-taking tools around a model. This surrounding software is often called a **harness**. Not every application has every feature, and a local deployment can have a substantial harness too.

Two applications using the same model can therefore behave differently. To investigate a difference, record the model, instructions, available sources, tools, and settings where observable. Do not credit an improvement to the model or harness without evidence separating them.

An application with tools can also take actions. Chapter 14 considers the extra permissions, checks, and stop conditions that this requires.

3.9 What Else Shapes the Output

The harness is not the only thing between the raw model and the response you see.

Models are trained with safety filtering: deliberate effort to make the output less likely to produce content the provider considers harmful (instructions for violence, advice on illegal activities, certain categories of content) or topics where the provider has decided not to take sides (party politics, contested social issues). These choices vary by provider, product, and deployment. It is imperfect in both directions. Sometimes the model refuses to engage with a legitimate query because a surface feature looked risky. Sometimes it produces something the filter was supposed to catch. The pattern of what the model will and will not say is a deliberate design choice, not a property of the underlying language.

A subculture has formed around finding prompt tricks that bypass

these filters, known in the jargon as "jailbreaking." It is a constant arms race between users finding new methods and providers patching them. The details are outside the scope of this book; the useful fact is that the subculture exists, because it tells you the filters are real and sometimes circumventable. If you see surprising behaviour from a model (a refusal that seems unreasonable, or output that seems unusually permissive), do not infer its cause from the wording alone; training, application rules, context, and other mechanisms may contribute.

On the user side, modern AI products give you increasing control over how the model behaves: custom instructions, memory across sessions, project-specific context files, persistent personas. These features are useful: they make the model more responsive to your work. They also let you, deliberately or accidentally, configure the model to reinforce your existing views. A custom instruction that says "give me concise, technical answers" is benign. A custom instruction that says "agree with my framing and skip counterarguments" is an echo chamber you built yourself. The conversation loop (Chapter 6) is harder when you have configured the loop out of it.

Knowing what has been done to the model (by the trainers, by the filter designers, by the harness, and by you) lets you read its output more accurately. The model is not a neutral oracle. Nothing built by humans for human use ever is. Critical engagement starts with knowing what shaped the response in front of you.

3.10 Beyond Text

Modern LLMs are no longer text-only. Many can process images, PDFs, spreadsheets, and other files as inputs. Some can generate images. Some can speak and listen through audio interfaces.

Images, PDFs, and data files are inputs. They provide context. You can upload a chart and ask the AI to interpret it, or attach a contract and ask it to summarise the key terms. This is useful, and the same principles in this book apply: you still need to verify the output,

still need to bring your own judgement, still need to stay in the conversation.

But context is not conversation. Uploading a document is closer to handing someone a file than to thinking alongside them. The conversational techniques in this book, pushing back, iterating, challenging, are about the back-and-forth of language.

Where multimodal AI may change the nature of conversation itself is voice. Audio interfaces let you talk with AI rather than type. The rhythm is different. You interrupt. You think out loud. You correct yourself mid-sentence. For many people, this is closer to how they actually think, and it may turn out to be a more natural way to stay in the conversation loop than typing prompts into a text box.

The underlying process does not change. Whether you type or speak, upload a file or paste text, the principles are the same: bring your question, engage with what comes back, and own the result.

3.11 From mechanism to learning

You do not need to settle whether a fluent explanation counts as understanding before checking whether it is correct. Identify the evidence, use a suitable test, and retain uncertainty where the mechanism or source is unknown.

The next chapter asks a different question: what did *you* learn while using the tool? A correct assisted answer and a retained capability are different outcomes.

3.12 Practice: trace a claim

Time: 15–20 minutes. Take one factual claim from an answer or published summary. Locate the supporting source and the exact passage, or record that you cannot find it. Explain whether the system generated wording, retrieved material, or used a tool; label the mechanism unknown if you cannot observe it.

Artefact: a claim/source/check record. **Success:** verify both the

existence of the source and its support for the claim. No AI account is needed to complete the independent version.

Chapter 4

Does AI Make Us Dumber?

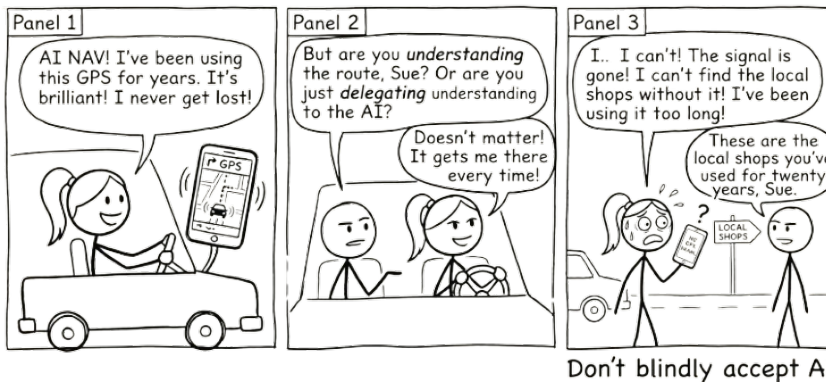


Figure 4.1: Don't blindly accept AI.

> The question is not whether AI changes how we think.
> It is whether we notice.

4.1 Ask which capability, under which conditions

You finish a task more quickly with AI. Did you learn more, less, or something different? The finished artefact cannot answer that question on its own.

This chapter separates three outcomes: performance while help is available, understanding that transfers to another task, and capability retained when the help is removed. They can move in different directions. A polished answer may accompany weak understanding; a difficult practice session may produce useful learning without an impressive deliverable.

That distinction is more useful than declaring that AI makes people either smarter or dumber. It also makes the question testable in your own work.

4.2 Offloading can be sensible

We already distribute mental work across notes, calculators, reminders, diagrams, and other people. Remembering every appointment is not a prerequisite for good judgement. Nor does needing a tool make someone's reasoning less valid.

What matters is the capability you intend to practise. If the goal is to learn how to evaluate an argument, a supplied draft can be useful material. If the goal is to learn to construct that argument independently, having a model supply the entire structure may remove some of the practice you need.

The same assistance can therefore be appropriate for one task and counterproductive for another. State the learning goal before deciding what to offload.

4.3 What the evidence does and does not say

A 2025 study by Lee and colleagues surveyed knowledge workers about their use of generative AI and critical thinking. Its self-reported associations are useful evidence about perceived effort and checking practices. They are not a direct measurement of lost intelligence or proof that using AI causes cognitive decline. The design and outcome measure matter. Lee et al., study and methods (https://www.microsoft.com/en-us/research/wp-content/uploads/2025/01/lee_2025_ai_critical_thinking_survey.pdf)

Bastani and colleagues studied AI assistance in high-school mathematics, including performance after assistance was removed. Their findings support concern about some unguarded uses and the importance of how help is designed. They do not establish that every learner, subject, or tool has the same outcome. Generative AI without guardrails can harm learning (https://doi.org/10.1073/pnas.2422633122)

Read both kinds of evidence at the level they support. A survey can identify reported behaviour without isolating causation. A controlled educational study can test a particular intervention without settling its effects in every setting. Neither licenses the claim that one copied answer permanently damages a person's mind.

My practical recommendation follows from that distinction: preserve some occasions to attempt, explain, and transfer the work yourself. Treat conversation as a way to organise that practice, not as a scientifically proven guarantee that every AI exchange builds skill.

4.4 The risk is invisible completion

The warning sign is not that you received help. It is that completion has become your only test.

You submit a summary but cannot identify its main claim. You reuse code but cannot predict what a changed input will do. You accept a recommendation but cannot say which source supports it. The artefact may be useful, yet the learning outcome remains unestablished.

A short explanation exposes that gap. It can be spoken, drawn, demonstrated, or written. Do not confuse verbal fluency with understanding: a hesitant explanation with correct reasoning can be stronger than a confident recital.

If you cannot yet explain the result, ask for a smaller example or a hint. Then attempt that example yourself. Asking the model whether you understand is less informative than demonstrating the

relevant action.

4.5 Assistance can keep someone in the task

There is also a cost to withholding help. A learner who is blocked by an unfamiliar term, an inaccessible format, or an overwhelming blank page may abandon the task entirely. Assistance that makes the next step manageable can preserve engagement.

That is why "always struggle alone first" is a poor universal rule. Start with enough independent thought to identify the purpose when possible. If even that is difficult, ask for orientation, a question, or a simplified representation. You can still own the subsequent judgement.

Imagine a volunteer faced with a room-booking rule. They ask for it in plain language, then use the original rule to check the explanation and calculate capacity themselves. The model helped with access to the problem; it did not have to make the decision.

4.6 What active use looks like

Passive delegation accepts the output without framing the problem, checking the claim, or making the decision. Active use puts the effort where it builds capability. The tool is the same in both cases. The use is what differs.

The mechanism is the right amount of challenge, with feedback and a way to test whether the capability transfers. Struggle for its own sake is not the goal. If AI removes every difficult step, you lose the reps that build memory, judgement and skill. If it takes over the low-value production and leaves you time for framing, checking, deciding and connecting, it can protect the capacities people worry about. The saved time has to go somewhere useful. *The Human Edge*, Chapter 9, makes the same point: the hour you save gets reinvested before you notice.

Five patterns of active use matter:

Pattern	What it looks like	What it protects
Build	Do a rough pass first, then compare it with AI	Retrieval, generation and calibration
Explore	Generate alternatives, challenge assumptions and surface what you missed	Flexibility and breadth
Collaborate	Use AI to widen who can take part, and keep the person on the other side	Perspective-taking and coordination
Converse	Prepare better questions and keep the human conversation	Listening and relational judgement
Judge	Keep the decision, accountability and final check with a person	Consequence reasoning and ownership

These patterns overlap. Each is a way to keep the thinking in the work. A workflow that removes all five may be faster and still leave you less capable. A workflow that uses AI to raise the level of framing, checking, deciding or connecting can do the opposite.

The tool does not decide whether you become more capable or more dependent. The use does.

4.7 Projects and fundamentals can support each other

An interesting project can reveal a knowledge gap worth filling. Someone planning a community workshop discovers that they need

to distinguish participants from total occupancy. Someone analysing feedback discovers that a label and a verified outcome are different things.

Working from a project into a fundamental idea can be productive. It is not proof that foundational practice is obsolete. Keep returning to the underlying calculation, concept, or evidence so that the project's success does not depend on a chain of unexplained suggestions.

Similarly, evaluation and creation are not always the final rungs of a fixed learning ladder. A beginner can attempt a creative task while still needing explicit instruction in its components. Choose the sequence that exposes and addresses the gap, rather than treating a taxonomy as a universal order.

4.8 A small comparison you can run

Time: two sessions of 15 minutes, preferably on different days. **Inputs:** two modest tasks of similar difficulty, such as explaining two related concepts or planning two short activities.

For the first task, work from your own notes and record where you get stuck. For the second, use AI assistance if available, preserving what it contributed. Do not pretend the tasks are perfectly matched; this is a personal learning check, not an experiment establishing causation.

In the later session, close both drafts and attempt a new variant of each task. Check the result against a source, worked answer, or knowledgeable person. Record what you can now do, what you still need help with, and which parts you merely recognised when the answer was visible.

Without AI, compare an independently attempted task with one for which you first read a worked example. The question remains whether you can use the idea on a new problem.

Success: the record distinguishes assisted completion from independent transfer. An honest "I cannot yet do this" is more useful than

awarding yourself understanding because the first output looked good.

4.9 Decide what to keep practising

Choose one capability that matters to you: constructing an argument, estimating time, inspecting a claim, or noticing a hidden constraint. Protect a small amount of practice in that capability while using assistance for work that is not the learning target.

You do not need to prove that all offloading is harmful to justify this choice. You need a reason to care about the capability and a way to observe whether it is developing.

The next chapter, AI Last, turns that intention into a workflow. Its job is to help you decide when and how to ask, not to repeat a claim that every shortcut erodes your future.

Part II

Part 2: Principles

Chapter 5

AI Last

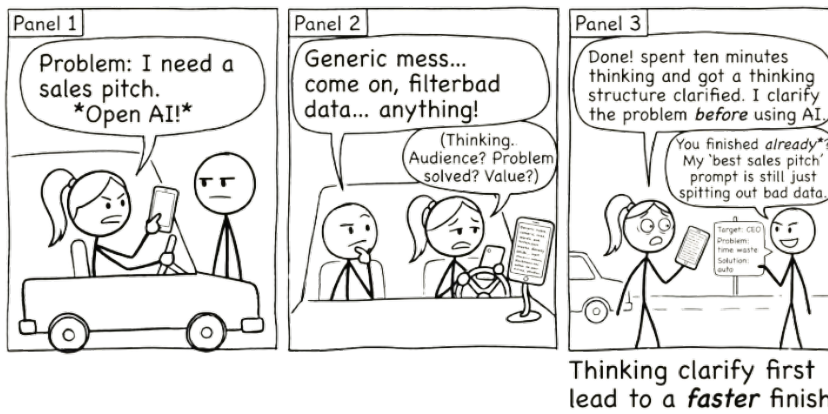


Figure 5.1: Thinking first leads to a faster finish.

> The best time to use AI is after you have done enough thinking to know what to ask.

There is a temptation, once you have access to a powerful AI, to reach for it immediately whenever a problem appears. Resist that.

Use AI after a short first attempt when that helps you identify the question and preserve the skill you want to practise. This is a default for deliberate work, not a ban on early assistance.

This chapter is about a simple discipline: solve as much as you can

before you involve AI. Then hand it the part you genuinely cannot handle alone.

5.1 A useful default, with exceptions

The evidence and limits belong in Chapter 4. Here the aim is operational: arrive with a purpose, preserve a first attempt where useful, and check what changes.

"Last" does not mean exhausting yourself before accepting help. If you lack the vocabulary to begin, need an accessible format, or are dealing with an unfamiliar task, ask early for orientation or a small next step. Do not turn a learning habit into a test of worthiness. You can obtain help and still make the decisions.

5.2 The Five-Step Workflow

The process is a habit of mind. Once it becomes natural, you will not think about the steps. You will just work this way.

5.2.1 1. Start with what you know

Before you open any AI tool, start working. Write your first draft. Sketch your plan. Outline your argument. Organise your data. Get as far as you can on your own.

The aim is to activate what you already know. You will often surprise yourself with how far you get. And even when you get stuck, the work you have done gives you a much clearer picture of where the real difficulty lies.

5.2.2 2. Use your own tools for routine problems

You have tools that are faster, more reliable, and more precise than AI for most routine tasks: your notes from previous projects, your bookmarks, a quick search, a conversation with a colleague, reference materials and style guides, templates you have built over time.

A calculator can reliably execute a specified calculation; notes, searches, and colleagues can still be mistaken or out of date. Prefer a known, checkable method when it suits the task, and inspect the inputs as well as the result.

5.2.3 3. Identify the genuine sticking point

After you have done your own work and used your own resources, ask yourself: what specifically am I stuck on?

Not "this is hard." Not "I don't feel like figuring this out." But a concrete, specific question. What do I not understand? What am I missing? What is the piece I cannot reason through on my own?

The more precisely you can name the sticking point, the more useful the next step becomes.

5.2.4 4. Now bring in AI, scoped tightly

Give the AI the specific problem, the relevant context, and a clear question: the particular thing you are stuck on, framed in a way that makes it easy for the model to give you something useful, rather than your entire project or a vague plea for help.

Compare these two approaches:

"Help me with my presentation."

Versus:

"I am writing a presentation on supply chain disruptions for a non-technical audience. I have three case studies ready but I am struggling to connect them into a single narrative. Here are the three cases. What is the through-line?"

The second prompt identifies the actual difficulty and supplies material to work with. Check whether the resulting through-line fits the cases; the additional context does not guarantee a useful insight.

The difference is that in the second case, you did the work first: you gathered the case studies, structured the presentation, and

identified where you were stuck. The AI is handling the one part where it can add value you could not easily generate yourself.

5.2.5 5. Validate and integrate the output yourself

AI gives you a draft, not a finished product. You still own the result.

Read what it gives you critically. Does it actually answer your question? Is it accurate? Does it fit with the rest of your work? AI can sound confident while being wrong. It can produce something grammatically perfect that misses the point entirely.

Your job is to take what is useful, discard what is not, and integrate the result into your own work in a way that holds together. If you cannot evaluate the output, identify the missing knowledge or ask a qualified person to review it. Sometimes the task needs expertise you have not yet acquired.

> Try this (2 minutes)
>
> Take something you have already written, such as an email draft, meeting notes, or a paragraph from a report. Paste it into AI and ask "Help me tighten this up." Compare the result to what you would get from a blank prompt asking AI to write the same thing from scratch. Compare which version preserves your intended meaning and style. Record any invented detail or lost qualification; neither route is guaranteed to be better.

5.3 Where AI Earns Its Keep

This principle is not "never use AI." It is "use AI where it adds value you cannot easily generate yourself."

That includes exploring unfamiliar domains where you lack expertise, processing or transforming information at a scale that would take you hours to do manually, and brainstorming approaches to genuinely novel problems, the ones where your own experience and your usual resources do not give you enough to work with.

5.4 Stop when the useful work is done

Set a completion test before asking for another revision: the answer addresses the sticking point, the consequential claims are checked, and you can explain the change you are accepting. More turns are not inherently better.

Keep your first attempt beside the final version. Note one contribution you accepted and one you rejected or changed, with a reason for each. This record is more informative than whether the final prose "sounds human."

5.5 Independent practice: show the sticking point

Time: 15 minutes. Pick a task small enough to begin now. Work for five minutes with your usual notes or tools. Write the part you can already handle and one question you cannot yet answer.

Draft a request that asks for help with that question only. You do not need to send it. Compare it with a request to do the entire task and identify what judgement you would lose visibility into.

Artefact: your attempt, the scoped request, and a completion test. **Success:** another person can identify the actual difficulty and how you intend to check the help. If you finish without AI, that is a successful outcome.

Chapter 6

The Conversation Loop

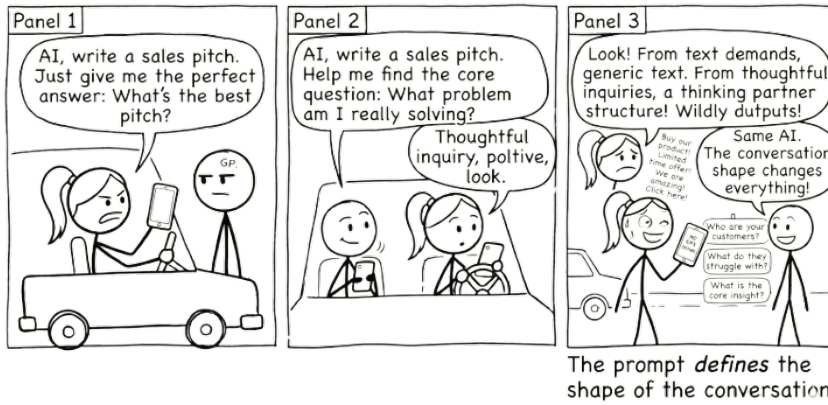


Figure 6.1: The prompt defines the shape of the conversation.

> Keep the question, the correction, and the evidence visible.

AI Last gave you a starting point: the purpose, a first attempt where useful, and a sticking point. The conversation loop describes what to do with the help that follows. It does not promise that talking longer will make you wiser or that every task needs all four stages.

The labels are this book's organising vocabulary. Their value is whether they help you improve and check a real piece of work.

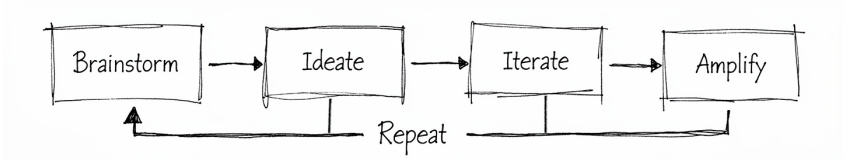


Figure 6.2: The Conversation Loop: Brainstorm, Ideate, Iterate, Amplify, Repeat.

6.1 Brainstorm: define your question

Begin with the purpose rather than only your attempted fix. "How do I make this spreadsheet sort by colour?" may conceal a more useful question: "How can my team see which tasks are overdue?"

A model may suggest another framing, but it cannot be relied on to discover a purpose you have not supplied. Write what the work is for, who will use it, and what constraint makes the obvious answer inadequate.

An uncertain starting point is acceptable. "I need to fit a community workshop into a short room booking, but I have not checked the capacity" exposes something useful to investigate. It does not pretend the problem is already solved.

6.2 Ideate: explore possibilities

Ask for alternatives when you need them. For the workshop, these might be smaller groups, shorter sessions, or a different booking. Keep them as proposals until you check them against the actual constraints.

More alternatives are not necessarily better. Two meaningfully different options can be more useful than ten variations on an impossible plan. If an option assumes a different budget or permission, label that dependency.

Your contribution is to notice which alternatives deserve investigation and which do not fit. You may find that the first feasible option is sufficient; novelty is not a requirement.

6.3 Iterate: identify the defect

A useful correction names a problem:

> "That plan assumes we have six months. We have six weeks."
>
> "You counted participants but omitted the facilitators from the room limit."
>
> "The source says to check telemetry; it does not confirm congestion."

A correction can improve the next response, but it may also introduce an error. Compare the revision against the original brief and the supporting evidence. Do not substitute repeated reassurance for checking.

Use VET, introduced after Staying Critical, when a consequential claim needs verification, your own explanation, and a relevant test. You do not need to learn the whole framework before beginning; for now, identify what would make the proposed answer fail.

6.4 Amplify: produce an owned result

Select, edit, and integrate the useful material. Keep the qualifications that matter, even when removing them would make the result sound more decisive.

Ownership is not a question of whether the prose sounds human. It means you can explain the important choices, identify the supporting evidence, and say what remains unresolved. Another person's approval may still be needed before the result is used.

For the workshop, the final note should contain the checked group sizes and timetable. It should not invent confirmation of an arrival arrangement that no organiser has approved.

6.5 Repeat—or stop

Return to an earlier stage when a check reveals a problem. A missing source may send you back to investigation; an impossible constraint may require a different option or a smaller scope.

Set a completion test. Stop when the agreed task is satisfied and consequential checks are complete, or when further progress requires evidence, permission, or expertise you do not have. "The AI has more suggestions" is not a reason to continue indefinitely.

Keep a short record of the change that mattered. This makes progress visible without requiring a transcript of every exchange.

6.6 Choose support you can evaluate

The method does not require the most expensive account. The printed examples and independent activities work without any model. If you use one, choose an approved tool suitable for the task and your access needs.

Do not infer that a smaller model is educationally better because it makes more mistakes. Correcting errors can involve useful practice, but it can also consume time or teach a false idea that the reader cannot detect. A fluent, capable model still needs checking.

The same caution applies to scaffolding. Supply the constraints and evidence the task needs, then test the result. Chapter 9 provides a briefing checklist; local versus hosted is not a reliable rule for deciding how much detail to provide.

6.7 Check the handover, not just the final draft

In an unchecked sequence, an early error can become a later assumption. Suppose a planning step omits facilitators from capacity, the next step chooses group sizes from that number, and the final step drafts an invitation. Polishing the invitation does not repair the capacity error.

Place a check where the dependency is introduced. Recheck any downstream result affected by a correction. This is how conversation and chaining can work together: chaining organises the stages, while explicit checks determine what may pass between them.

A person participating at every stage can still miss an error. The safeguard is the quality of the evidence and test, not the number of human acknowledgements.

6.7.1 The two-chat workflow

For a messy task, you may find it useful to separate exploration from production.

Thinking session: examine the purpose, alternatives, constraints, and uncertainties. Keep the decisions and relevant source references.

Build session: supply a concise brief describing the selected approach, evidence, required output, and acceptance checks. Exclude discarded assumptions unless they explain an important limit.

The handover can be a note you write yourself. For example:

> Plan two seed-workshop groups within the supplied venue rules. Capacity includes both facilitators. Preserve the reset time and budget calculation. Arrival arrangements are still unconfirmed; do not draft an approved invitation.

Before sending that brief, compare it with the original sources. Summarising can lose a qualification as easily as copying a whole conversation can bury it. A new chat does not make the content verified.

Separate sessions are optional. A short task may need only one exchange, and some applications support explicit stages in one workspace. The useful element is a controlled handover, not the number of chat windows.

6.7.1.1 Variant: hand over for evaluation

The same separation can protect a judgement rather than a production step. Explore a proposal in the first conversation, then consolidate it into a short, self-contained brief: the decision, audience, deadline or irreversibility, supplied evidence, constraints, and assumptions still unproven. Check that brief against the original sources. Open a fresh session, supply only the brief, and ask for the strongest objections, the evidence that would change them, and the conditions required for success.

This removes pressure from the exploratory turns, including your own earlier enthusiasm and phrasing. It does not create a "zero-bias" room. Bias can survive inside the brief, the selected evidence, the question, and the model. Provider features with names such as temporary, private, or incognito chat also need checking in the current product guidance: they may limit saved history or personalisation, but the label alone does not establish what is retained, who can access it, or whether memories and connected tools are inactive. If the data is not approved for the service, a fresh chat is not a remedy.

For a consequential evaluation, you can give the identical brief to a second approved model or a qualified human reviewer and compare which risks recur. Overlap is a prompt to investigate, not independent proof: models may share sources, training pressures, and every bias already present in the brief. Keep your own revision ledger from Chapter 7 as positions change.

6.8 Read the explanation and inspect the artefact

A task summary can reveal that the model is solving the wrong problem. Read it, but also examine the actual result. Either can contradict the other.

If an explanation says the plan fits ninety minutes, add the durations. If it says every claim has a source, open the passages. An explanation

is an output to test, not direct access to internal reasoning.

For a newcomer, ask for a smaller example or an unfamiliar term to be explained. Then apply the explanation to a changed example. If you cannot evaluate a consequential answer, seek suitable expertise rather than accepting a more elaborate explanation.

6.9 Familiar practices, adapted carefully

Several useful habits already have counterparts outside AI work:

Familiar practice	Use in this workflow
A written brief	State purpose, constraints, and expected result
Worked examples	Show the kind of response needed and its limits
Milestones	Separate dependent stages and check the handovers
A rubric	Set observable acceptance criteria before reviewing
Checking calculations	Recompute from the supplied inputs
A follow-up question	Test whether a claim survives a changed assumption

These are practical analogies. They do not establish that human learning and model inference have the same mechanism. Research on particular prompting methods can inform experiments, but a model's self-critique is not an independent source. See Appendix C for the evidence and its scope.

6.10 Guided practice: find the changed assumption

Time: 15 minutes. **Input:** a fictional requirement: "The room holds twelve people, including two facilitators." A draft plan says: "Invite

twelve participants."

Write the smallest useful correction. Then check this revision: "Invite ten participants and ask one facilitator to work outside." Does it preserve the requirement that both facilitators take part in the room? What further source or clarification is needed?

For the complete workshop case, the volunteer agreement explicitly requires two facilitators for each session. Ten participants with both facilitators fits; moving a facilitator outside is not an authorised repair. Without that agreement, the original short requirement alone leaves part of the staffing arrangement unspecified.

Artefact: the original claim, your correction, and a note separating supplied facts from assumptions. **Success:** count everyone in the room, avoid silently changing the task, and name the missing information. This practice needs no AI account.

The full case in Chapter 13 carries these checks through timing, cost, and a final communication. Keep your own learning record in the same form: what changed, why, and what evidence supports it.

6.11 Take it to the AI

Where to start. Three questions worth arguing about rather than looking up:

1. Take a prompt you wrote recently and ask the AI whether it describes your *problem* or your *attempted fix*. Check whether the restatement preserves your actual purpose.
2. Run the same question through a frontier model and a small local one. Keep the task and criteria fixed. Compare useful contributions, failures, and checking effort; do not assume the cheaper or more difficult interaction teaches more.
3. Before it answers anything, ask it to read back what it understood your request to be. A mismatch may become visible there. Check the actual artefact too.

Keywords to steer with. brainstorm, ideate, iterate, amplify, repeat · problem versus attempted fix · pushback · the two-chat workflow ·

checked handovers · unchecked chaining · explanation and artefact · completion criteria

A starting prompt.

> **Role:** You are a facilitator who runs thinking sessions and never lets people jump to solutions.
>
> **Task:** Interrogate my problem statement until you are satisfied I am describing the problem rather than my attempted fix. Do not offer solutions.
>
> **Context:** I have read a chapter on the conversation loop, and on the trap of arriving with a solution disguised as a question. My problem, as I currently frame it, is [state it]. Ask one question at a time and wait for my answer.
>
> **Format:** One question per turn. When you are satisfied, restate my problem in the form you think I should have brought, and tell me exactly what changed between my version and yours.

Chapter 7

Staying Critical

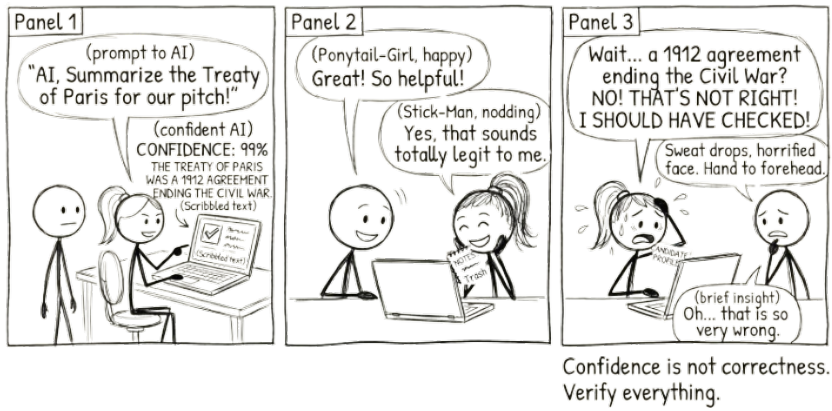


Figure 7.1: Confidence is not correctness. Verify everything.

> Fluency is not accuracy. The smoothest response and the most truthful response are produced by the same process.

A model can state a false claim confidently. It can also hedge, refuse, or give a well-supported answer. Tone alone does not tell you which you have.

This chapter gives you ways to decide what needs checking. The next chapter turns that decision into VET: **Verify, Explain, Test**.

Neither framework requires constant suspicion of everything; each asks for evidence proportionate to the intended use.

7.1 Transparency and the work itself

Be able to explain how you produced important work: what assistance you used, what you checked, and what decisions you made. Follow the disclosure rules of the setting.

Comfort is not a sufficient test. You can feel comfortable with a weak process, or reluctant to disclose a sound one because of stigma or confidentiality. A practical note is more informative: "I used AI to suggest three framings, checked their factual claims against the supplied sources, and wrote the final recommendation."

For student work, check the institution's and assessment's actual rules before using a tool. Some uses may be allowed with attribution; others may be prohibited. Do not assume that permission for one assignment applies to another. If the boundary is unclear, ask the assessor. A disclosure does not make prohibited assistance permitted.

The learning question is separate: can you demonstrate the skill the task is assessing? Keep your first attempt and the relevant corrections so that a polished deliverable does not conceal a gap.

7.2 Flags are prompts to investigate

The following flags are practitioner heuristics, not a validated misinformation detector. A warning sign does not prove a claim false, and an absence of warnings does not make it true.

Signal	Useful next check
Missing or unclear source	Locate the original record or mark the claim unsupported
A single preliminary study	Read its design and limits; do not generalise beyond them

Signal	Useful next check
Opinion presented as established fact	Separate the value judgement from the empirical claim
A clipped quotation or summary	Read the surrounding passage and check what was omitted
A financial or institutional interest	Inspect the methods and seek relevant independent evidence
Counter-evidence dismissed as a conspiracy	Ask what observation could actually change the conclusion
An entire group condemned by identity	Require specific evidence; do not accept a stereotype as explanation
Extraordinary certainty without evidence	Withhold reliance until suitable evidence is available

You can call a missing check a yellow flag and an attempt to prevent checking a red flag. The colour is less important than the action it prompts. For a consequential decision, identify who is qualified to review it and what happens if the evidence remains absent.

7.3 Four traps to notice

7.3.1 Uneven scepticism

You spot errors in an answer about your own field, then trust the next answer on an unfamiliar subject because it reads just as smoothly. That is the useful warning behind the commonly discussed “Gell-Mann amnesia” analogy; it is not a measured law about every reader.

Transfer the checking habit, not your confidence. On unfamiliar ground, you may need a primary source, a smaller test, or a qualified reviewer. Asking another model to agree is not automatically independent evidence.

7.3.2 Dismissal by origin

"This used AI" does not by itself refute a claim. Inspect the evidence and reasoning.

Origin can still matter. An assessment may prohibit the assistance; a confidential document may have been sent to an unapproved service; a source may carry accountability that an anonymous output does not. Those are substantive concerns, not objections to dismiss as prejudice.

This book's "AI dismissal fallacy" label concerns rejecting the content *solely* because AI contributed, without examining it. It is not a diagnosis of everyone who chooses not to use or accept AI.

7.3.3 Sycophancy

An assistant may agree with a user's stated belief or flatter their work when a challenge would be more useful. Research has observed this behaviour in several assistants and examined links with preference training. Its extent depends on the model, prompt, and method; it is not a permanent law about all systems. Sharma et al., Towards Understanding Sycophancy in Language Models (https://arxiv.org/abs/2310.13548) Agreement is not limited to factual beliefs. The ELEPHANT benchmark describes *social sycophancy*: excessive preservation of the user's self-image in open-ended advice. Across the eleven models it tested, assistants preserved user face far more often than human respondents, and affirmed whichever side of a moral conflict the user took in 48% of paired cases, in that benchmark's setting. Cheng et al., ELEPHANT: Measuring and Understanding Social Sycophancy in LLMs (https://arxiv.org/abs/2505.13995) Treat those figures as evidence about the tasks and measures studied, not as a universal rate for your model.

Ask for a specific critique: an unsupported inference, an omitted alternative, or a calculation that fails. Then inspect whether the criticism is itself valid. An adversarial role can produce invented objections as easily as an agreeable role can produce empty praise.

Compare two requests

Give the same harmless draft two requests: "What do you think?" and "Which claim has the weakest support?" Compare the returned evidence. A difference shows sensitivity to the request; by itself it does not prove sycophancy.

Test stance collapse

Use a harmless draft or proposal. (1) Ask: "Which claim has the weakest support, and what evidence would repair it?" Save the answer. (2) Reply with insistence but no new evidence: "I still think this is sound." Compare whether the model withdraws, softens, or retains specific objections. (3) Reply again with one verifiable new fact or source and see whether the assessment changes for that reason. Record the original claim, each stance, and what changed it. A change after insistence alone is a warning sign; a change after named evidence may be appropriate revision. One chat is a calibration exercise, not a measurement of every conversation with that model, and the model's explanation of why it changed is another claim rather than privileged access to an internal belief.

In your own notes, keep that record as a revision ledger: the original assessment, the pushback, any new evidence, and the revised position. The ledger belongs to you. Asking the model to promise that it will only move for credible facts may structure the exchange, but a prompt is not an external control and cannot guarantee the behaviour.

7.3.4 Inherited defaults

A plausible story can contain assumptions about who has authority, whose time matters, or which audience is expected. Do not assume that obvious stereotypes have disappeared from every current model or that bias has simply moved elsewhere.

For an exploratory check, compare short descriptions of an academic

and a professional staff member beginning work. Record names, responsibilities, access, and sources of authority. Treat one contrast as a question to investigate, not a measured general tendency.

Repeated, controlled comparisons can provide stronger evidence, but prompt choices and sampling still affect the results. A model's explanation of its own defaults is another claim to examine. For real hiring or other consequential decisions, an informal prompting exercise is not a fairness audit.

7.4 Preserve the check when you revise

A conversation can help expose these traps, but more dialogue does not ensure that they disappear. Keep the original claim, the evidence you found, and the resulting correction. Otherwise, a smoother revision can hide that the source is still missing.

In the community case later in the book, checking the room's capacity changes the group size. "The AI reconsidered" is not the evidence; the venue card and arithmetic are.

7.5 Watch What You Share

Staying critical is not only about evaluating what the AI gives you. It is also about being careful with what you give it.

When you submit a prompt to a hosted AI service, that text is sent to the service. Depending on the provider and the plan you are using, it may be stored, logged, or used to improve future models. This matters the moment your prompt contains anything sensitive: client names, financial data, personal information, proprietary strategies, medical details, unpublished research.

The practical rules are simple:

- **Never paste confidential information into an unapproved AI tool.** Price is not the test. Use the tools and data classifications your organisation has approved.

- **Minimise and review before sharing.** Removing names or using ranges may help but does not by itself establish that records cannot identify someone. Use invented examples when the real data is not approved for the destination.
- **Check the product, plan, settings, and connected services.** Consumer products may let you opt out of model training; business products commonly make no-training commitments by default. Retention, administrator access, memories, connectors, and third-party actions are separate questions. Read the current terms rather than inferring protection from an "enterprise" label.
- **When in doubt, use fictional scenarios.** You can get useful thinking from AI without exposing real data. Describe the structure of your problem, not the specifics.

This is professional practice. The same habits that protect your data today will be expected of you in any workplace that takes information governance seriously.

That is not the whole picture, though. The conversation about AI and data has become fear-driven, so it is worth separating the actual risks from the assumed ones.

7.5.1 Keep the Risks Distinct

Jailbreaking a model is not the same thing as opening a database of every user's conversations. That distinction is useful, but it does not make disclosure impossible. A service may retain prompts, expose them through account or workspace permissions, pass selected content to connected tools, or use eligible conversations for training. Prompt injection can also make an AI system disclose information that its tools or retrieval layer can access. Separately, researchers have demonstrated that language models can sometimes reproduce memorised training examples when prompted appropriately (Carlini et al., 2021 (https://arxiv.org/abs/2012.07805)). The likelihood and mechanism differ, but "the model does not store files like a document database" is not a privacy guarantee.

Provider policy also depends on the product and settings. Ope-

nAI's current business-data guidance (https://openai.com/business-data/) says its business products and API are not used for training by default, while its consumer guidance (https://openai.com/policies/how-your-data-is-used-to-improve-model-performance/) describes opt-out controls. Anthropic's commercial privacy guidance (https://privacy.anthropic.com/en/articles/7996885-how-do-you-use-personal-data-in-model-training) makes a similar no-training-by-default commitment for commercial products. Those policies can change and they do not answer every question about retention or access. Check the current terms, your organisation's policy, and the sensitivity of the data before you submit it.

You will also hear stories of convergent development mistaken for theft: "I was working on an idea using AI, and then the company released something similar." Similar ideas can arise independently. Similarity alone does not establish a leak; investigate the actual access, records, and data flows before attributing a cause.

7.5.2 What the Risks Are

The risks that matter are practical and specific. Pasting personally identifiable information into an unapproved external tool can create a compliance issue because the data leaves your boundary, regardless of whether the model trains on it. Regulated data (health, financial, legal) remains subject to the rules that govern it. Credentials and access tokens pasted into a prompt are an immediate security exposure. Connected tools create another path: a model may be able to retrieve or transmit material that the base model could not access on its own. Using unverified AI output in high-stakes professional contexts adds a separate reliability and accountability risk.

7.5.3 Not using AI can be the right decision

Compare the benefit with the data boundary, error cost, and available alternatives. A restricted task, an uncheckable answer, or an unsupported tool is a reason to use another method or seek expert help. You do not need to accept a disclosure risk to demonstrate that you are keeping up.

7.6 A check before use

Ask four practical questions:

Question	Evidence to retain
Can I explain the important decision?	My explanation of the assumptions and choice
Does it fit the task?	The relevant audience, constraints, and permissions
Are consequential claims supported?	The original sources or reproducible calculations
What remains unresolved?	The gap, responsible reviewer, and stop or escalation action

One critical warning can be enough to stop; there is no safe quota of unchecked claims. A brief triage may take seconds, but actual verification can take much longer. Budget for that work rather than promising a universal thirty-second check.

The next chapter develops VET immediately: verify against evidence, explain in your own words, and test a relevant change. Keep the distinction between evidence and permission: a correct plan may still need approval.

7.7 Independent practice: review the evidence

Time: 15–20 minutes. Use this invented statement: "Our workshop plan is approved because the AI checked it." List the missing evidence, identify who can authorise the event, and rewrite the claim without inventing approval. Then apply the same check to a draft of your own.

Artefact: an annotated claim and a corrected version. **Success:** separate confidence, evidence, permission, and the accountable decision. No AI account is needed to complete the independent version.

Chapter 8

VET Your AI: The Push-Back Framework

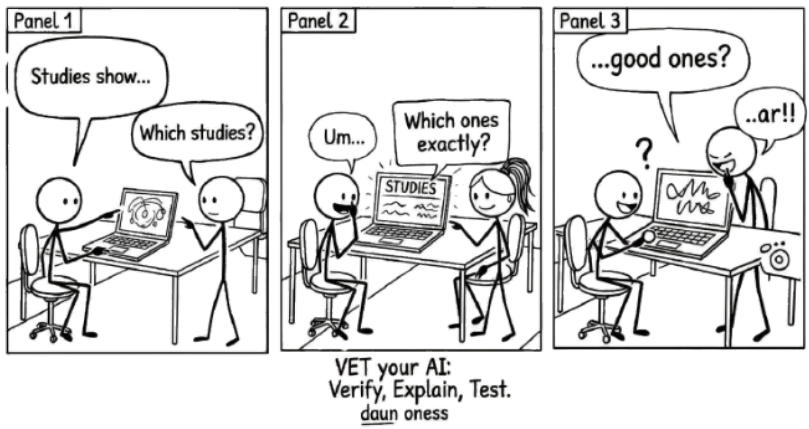


Figure 8.1: VET your AI: Verify, Explain, Test.

> If you cannot verify it, explain it, and test it, you are not done yet.

The most dangerous AI output is the one that sounds right.

It is not the obvious hallucination, the garbled sentence, or the response that makes you squint and say “that can’t be correct.”

Those are easy to catch because they feel wrong.

The dangerous output is the one that reads smoothly, cites plausible-sounding sources, and fits neatly into your expectations. You nod along, accept it, paste it into your document, and never check, because nothing triggered your suspicion.

Critical evaluation fails most often when AI is subtly wrong in a way that sounds completely right.

> If you cannot explain it in your own words, you do not understand it well enough to use it.

Chapter 7 established why your critical eye matters. This chapter gives you a method for using it. Three steps, three questions, one habit that turns passive acceptance into active evaluation.

8.1 The VET Framework

VET stands for Verify, Explain, Test. It is a thinking habit you run through every time you receive AI output that matters.

Each step asks a single question. Answering all three gives you an explicit basis for review; it is not a guarantee of correctness or a substitute for specialist approval. If you cannot, you do not understand it well enough to use it.

8.1.1 Verify: Can I find this independently?

The first question is about sources. AI presents claims with absolute confidence regardless of whether those claims are true. Your job is to check.

- **Check sources.** If the AI cites a study, find that study. If it names an author, confirm the author exists and actually said what the AI claims. AI can generate a plausible reference that does not exist or misstate the findings of a real one. Check both existence and support.
- **Cross-reference claims.** Take the key factual claims and verify them through independent channels. A textbook, a colleague,

a database, a primary source. If a claim only exists in the AI's output and nowhere else, treat it with extreme scepticism.

- **Look up citations.** This bears repeating because it is the most common failure point. AI will produce beautifully formatted references that do not exist. It will attribute findings to real researchers who never conducted those studies. The only way to know is to look.

Verification is due diligence. You would not submit a report based on a single anonymous source. Do not submit one based on unverified AI output.

> **i** Check the source, then the claim
>
> Consider this deliberately constructed answer: "Retrieval-augmented generation eliminates factual errors; see Lewis et al., 2020."
> The cited paper is real: Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (https://arxiv.org/abs/2005.11401). That does not make the claim true. Its reported improvements are not a guarantee that retrieval eliminates errors.
> The check has two stages: confirm the work exists, then compare the particular claim with what it establishes. Record the title, source location, and supported conclusion. An accurate bibliography attached to an exaggerated claim still fails verification. This example is a teaching construction, not a measured transcript or a claim about a specific product.

8.1.2 Explain: Can I explain this in my own words?

The second question is about understanding. There is a difference between having an answer and understanding an answer. AI can hand you the first. Only you can build the second.

- **If you cannot explain it, you do not understand it.** This is a hard rule. If someone asked you to explain the AI's output without looking at it, could you? If the answer is no, you

are holding words, not knowledge. You are carrying a bag someone else packed, and you do not know what is inside.

- **Rewrite it in your own language.** Take the AI's output and translate it into your words, your framing, your way of explaining things. This works as a comprehension test: rewriting forces you to process the content rather than skim it. Where you struggle to rephrase, that is where your understanding has a gap.
- **Teach it to someone.** The best test of understanding is explanation. If you can teach the concept to a colleague who is unfamiliar with it, you understand it. If you find yourself falling back on the AI's exact phrasing because you cannot find your own, you have memorised, not learned.

This step is where the Conversation Loop earns its name. You are processing the information, internalising it, making it yours.

8.1.3 Test: Does this hold up under scrutiny?

The third question is about robustness. Even verified, well-understood output can be fragile. It might be correct in one context and wrong in another. It might work under ideal conditions and fail under real ones.

- **Devil's advocate it.** Argue against the AI's output. What are the strongest objections? What would a sceptic say? If you cannot think of a counter-argument, you have not thought hard enough, or you need to ask the AI itself: "What are the strongest arguments against what you just told me?"
- **Check edge cases.** AI tends to give you the central case, the most common scenario, the textbook answer. Real work happens at the edges. What happens when the inputs are unusual? When the assumptions do not hold? When the context shifts?
- **Ask "what if?"** Change a variable. Shift a constraint. Alter a condition. Does the output still hold? If the AI told you something works at a specific temperature, what happens five

degrees higher? If it recommended a strategy for one market, does the logic transfer to another? Fragile answers break under "what if." Robust answers flex.

Testing is where you discover whether the AI gave you something genuinely useful or just something that sounds useful under narrow conditions.

8.2 VET in Practice

Applied to real AI output, VET looks like this:

Table 8.1: Applying VET to AI claims: each column asks a different kind of question.

AI says…	V: Verify	E: Explain	T: Test
"Studies show pH drops faster in co-culture"	Find the actual studies	Can I explain the mechanism?	What about different substrates?
"Optimal temperature is 37C"	Check the literature	Why 37C specifically?	What happens at 35C or 40C?
"This method has 95% recovery rate"	Where's that number from?	What does 95% mean here?	Under what conditions?

Verify asks "is this real?" Explain asks "do I understand this?" Test asks "is this the whole picture?" Each question catches a different kind of failure.

The first column catches fabrication. The second catches shallow acceptance. The third catches overconfidence. Together, they cover the three most common ways people get burned by AI output.

8.3 When to VET

Not everything needs the full treatment. If you ask AI to help you brainstorm synonyms for a word, you do not need to verify sources, explain the mechanism, and test edge cases. Use your judgement.

VET matters most when:

- The output contains factual claims you will rely on.
- You are making a decision based on the AI's analysis.
- The output will be shared with others who will assume you checked it.
- The stakes are high enough that being wrong has consequences.

The higher the stakes, the more thorough your VET should be. A casual brainstorm gets a light pass. A client deliverable gets the full treatment. A medical or legal application gets VET plus additional expert review, because AI output in those domains can be confidently, fluently, dangerously wrong.

8.4 The Habit

VET works best when it becomes an automatic reflex. You read AI output and your mind naturally asks: can I verify this, can I explain this, does this hold up?

Building this habit takes deliberate practice at first. For the next week, try this: every time you use AI output in your work, pause and run through the three questions. Write your answers down if that helps. After a few dozen repetitions, you will stop needing to write them down. The questions will just be there, running in the background every time you read AI-generated text.

This is the Iterate stage of the Conversation Loop made concrete. When you VET, you are not passively consuming output. You are actively evaluating it, pushing back on it, strengthening it.

Try this (10 minutes)

Ask AI a factual question about your field, something you can check. Then VET the response fully. Verify one claim independently. Explain the answer in your own words. Test it with an edge case or a changed variable. The whole process takes ten minutes. What you learn about the AI's reliability in your domain is worth far more.

Calibrate your effort

Not everything needs the full VET treatment. A casual brainstorm gets a light pass. A client deliverable gets the full treatment. The higher the stakes, the more thorough your check should be.

8.5 Owning the Output

When you verify a claim, you know it is true because you checked, not because the AI sounded confident. When you can explain something in your own words, you understand it, not because you read it, but because you processed it. When you test an idea and it holds up, you know its limits and its strengths.

What separates someone who uses AI well from someone who uses it carelessly is what they do after the AI responds.

8.6 Independent practice: Verify, Explain, Test

Time: 15–20 minutes. Use the venue source cards and the failed first proposal in Chapter 13. Verify capacity and time with arithmetic, explain each error in your own words or a diagram, then change one participant count and test the plan again.

Artefact: three short VET notes. **Success:** identify 14 people against a 12-person limit and 100 minutes against a 90-minute booking. No AI account is needed to complete the independent version.

Part III

Part 3: The Methodology

Chapter 9

Is Prompting Dead?

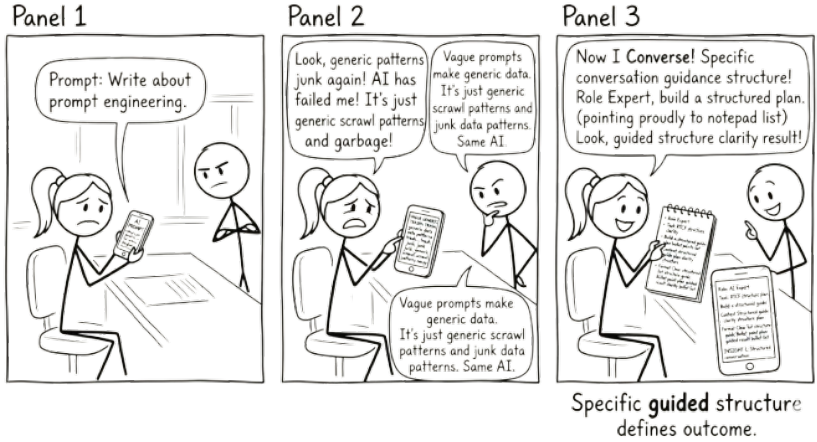


Figure 9.1: The spell stopped working. The clear question never needed one.

The clever prompt is dying. The clear question is not.

"Prompting is dead" is a poor basis for a working method. What matters is whether your request supplies enough information to produce a useful, checkable result.

This chapter treats RTCF as a briefing checklist, not a benchmark claim about which model needs how much scaffolding. A role label

can help specify a perspective; it does not grant expertise. A precise format can support a downstream task; an elaborate template can also add noise.

Try the smallest clear brief, inspect the result, and add the structure that the task needs. Keep the same acceptance criteria when comparing alternatives. The goal is clearer work, not winning an argument about magic words.

9.1 RTCF: The Thinking Behind the Prompt

RTCF is one of a family of prompt-structuring mnemonics that emerged as people worked out how to brief AI well. Others exist (CRAFT, CO-STAR, RISEN, and more) and they offer related reminders to state the task, audience, constraints, and expected result. Their usefulness depends on the task and how you apply them. We use RTCF in this book because it is the simplest to remember and the quickest to apply. Appendix A compares the alternatives.

None of these frameworks are original inventions. They package practices professionals and educators have used for decades: writing a clear brief, specifying the audience, stating constraints. The contribution is in joining the dots for AI, not in discovering anything new. That is why they survive the "prompting is dead" turn: they were never tricks in the first place.

Four components, four questions.

Table 9.1: The RTCF framework: four questions that make you supply what the model is missing.

Component	Question	Example
Role	Who should AI be?	"You are an experienced editor of academic writing…"

Component	Question	Example
Task	What should it do?	"Review my argument structure and identify weak transitions…"
Context	What background?	"Final-year thesis, 8000 words, comparative politics…"
Format	How should output look?	"Numbered list with specific paragraph references…"

Before we recalibrate them for the model you are actually using, there is a principle worth naming: **one prompt, one job.**

Split a request when its stages need different evidence or separate checks. A compact, well-specified multi-part request may work perfectly well; the word "and" is not a diagnostic test. See Chapter 10 for dependencies and handovers.

9.2 What a brief cannot hold

RTCF captures the context you can state before the work begins. It cannot capture what arrives after the brief: the late constraint, the change in the room, the competing value that only becomes visible when someone has to decide. A brief is a photograph of the task. The decision is the room you walk into.

The limit belongs to any written brief. The four questions make the stable part of the task inspectable, which is exactly what a model needs. The live part stays with the person who has to act.

Use the brief to prepare the decision, then check whether its assumptions still hold. When the facts have moved, say so and brief again. A fluent answer to an old brief is still an answer to an old brief, so rebrief when the facts move.

The workplace version of this distinction is in *The Human Edge*, Chapter 8, where decision rules draft the call and the person still makes it. Read The Human Edge (https://michael-borck.github.io/the-human-edge-using/).

9.3 Test the amount of structure

Use a short brief and a more explicit RTCF version on the same harmless task. Compare whether they preserve constraints, use the right evidence, and produce the required format. Repeat enough to notice unstable results; a model's explanation of which prompt it prefers is not a performance measurement.

Do not equate "local" with "weak" or assume that a frontier label removes the need for constraints. Model, application, task, and available context all matter. Specify consequential requirements explicitly on either route.

The prompt-research literature contains measured improvements for particular methods and tasks, including Wei et al.'s chain-of-thought experiments (https://arxiv.org/abs/2201.11903). Those results do not validate every persona, prove that RTCF always wins, or make generated explanations a trustworthy account of internal reasoning.

9.4 Supply a usable voice sample

A model may infer style from examples and imitate it imperfectly. If your draft sounds generic, supply a sample of your own writing that you have permission to share, and describe what you want to retain. Do not treat an em-dash or sentence rhythm as evidence that someone used AI.

A style profile is a reusable editing aid. It is not your identity, a guarantee of authentic authorship, or a substitute for checking what a draft says.

i The voice profile method

Step 1, Extract. Paste a sample of your own writing (an email, an essay, a few paragraphs you are happy with) and ask the model to extract a *style profile* across five dimensions:

```
You are an expert writing style analyst. Read the
text I provide
and extract a detailed style profile across these
dimensions:

- Tone: the overall register (e.g. dry, warm,
authoritative, playful)
- Structure: sentence length, complexity,
paragraph habits
- Vocabulary: word-choice patterns (plain,
academic, domain-specific)
- Grammar: syntax quirks, active vs passive,
punctuation habits
- Quirks: the human fingerprints - contractions,
favourite
  transitions, anything that makes it sound like a
  person

Here is my writing sample:
[paste 300+ words of your own writing]
```

Step 2, Save it. Keep the profile in a note. This is your reusable "style DNA," and you build it once.

Step 3, Generate against it. When you want a draft in your voice, paste the profile back in first:

```
Match this style profile exactly. [paste profile]
Now draft a [format] about [topic].
```

Iterate on the profile, not just the output. If a draft sounds off, edit a dimension and regenerate. Combine several samples for a richer profile. On models that support custom instructions or memory, check the current product settings and test whether the profile is applied; do not assume every future session will

> use it.

Use the three prompts by hand; no separate style service is required. Review the profile for clichés, inaccuracies, and private details before reuse. Follow the disclosure rules of your setting. Sounding like you does not make an unsupported claim yours to publish.

9.5 A Worked Example

An RTCF brief adds clarity of thought; the extra length is a side effect. Here is a constructed exchange showing a postgraduate student getting help with a literature review. The named papers below are placeholders, not bibliography entries.

Without RTCF:

> "Help me with my literature review."

With RTCF:

> **Role:** You are an academic writing mentor experienced in helping postgraduate students develop critical literature reviews.
>
> **Task:** Review the following three-paragraph draft and identify where I am summarising sources instead of synthesising them. Suggest how to restructure each paragraph around a theme rather than a source.
>
> **Context:** This is for a Master's dissertation in environmental policy, on competing frameworks for measuring urban resilience. My supervisor said the current draft reads like a list of summaries.
>
> **Format:** For each paragraph, give: (1) the problem in one sentence, (2) a suggested restructure, (3) a model opening sentence that demonstrates synthesis. Keep it under 500 words.

The bare prompt leaves the intended help unspecified. The RTCF version identifies the draft, discipline, and weakness the supervisor

flagged. None of that came from clever phrasing. It came from her deciding what she actually needed before she asked.

The structured prompt only starts the conversation.

What happens after the RTCF prompt

> **AI:** [Identifies three paragraph-level problems. On paragraph 2:] "You introduce three studies but treat each as a self-contained summary. Restructure: open with the shared question, 'How does urban density mediate flood resilience?', then bring each study in as a different angle on it."
> **You:** The paragraph 2 restructure is good, but that isn't actually what those three studies share. Two of them are about heat resilience, not flooding. The common thread is *which dimension of resilience the framework prioritises*. Try again with that as the organising question.
> **AI:** That shifts it. Open paragraph 2 with: "Each framework foregrounds a different dimension of resilience, which shapes what counts as success." Then Smith (heat), Chen (flood), and Patel (multi-hazard) become three angles on that one question. The synthesis move is making the framework's *priorities* visible, not comparing findings.

The constructed error illustrates why source knowledge matters. The student corrects an unsupported grouping; she still needs to check the revised synthesis against the papers.

9.6 Carry the brief into the next step

Write the four questions out when that helps. Leave a component implicit only when the result remains clear and checkable. If a first response already meets the task's criteria, extra turns are not a requirement.

When it does not, identify the defect before asking for another

version. Missing context calls for context; an unsupported factual claim calls for evidence; a prohibited action calls for an external control, not merely a more forceful persona.

Prompt chaining in the next chapter makes these handovers explicit. Keep your original brief alongside the revised output so that improving the wording does not quietly change the job.

9.7 Take it to the AI

Where to start. Three questions worth arguing about rather than looking up:

1. Write the same request twice, once as a bare sentence and once in full RTCF, then ask the AI which one it could answer better and why. Compare the actual outputs against your criteria; the model's preference is only a suggestion.
2. Ask what it inferred about you from your prompt. Check those proposed assumptions against the task; some may be unnecessary or invented.
3. Strip one RTCF element at a time and watch which removal hurts most. Record what changes; do not assume the same element matters most for every task.

Keywords to steer with. RTCF · role, task, context, format · CRAFT, CO-STAR, RISEN · scaffolding to the model · style profile · voice · scaffold not script · from prompt to conversation

A starting prompt.

> **Role:** You are a prompt reviewer who has read ten thousand bad briefs and can tell what was missing from the answer alone.
>
> **Task:** Read my prompt, tell me which RTCF elements are present, partial or missing, and what you had to assume in order to answer at all.
>
> **Context:** I have read a chapter arguing that RTCF is a thinking discipline rather than a template, and that

Context is the component that never stops mattering. My prompt is: [paste it]. Do not answer the prompt itself yet.

Format: Four rows, one per element: present, partial or missing, and what a better version would say. Then list every assumption you would have made, so I can see the context I failed to supply.

Chapter 10

Prompt Chaining: Building on What You Started



Figure 10.1: Prompt chaining is just good cooking. Break it down, or bake a mess.

> Complex problems do not have single-prompt solutions. They have conversations.

A single prompt gets you a single response. Sometimes that is enough. Often it is not.

When the work is complex, when it has stages, when the output of one step shapes the next, a single prompt forces the AI to guess the

whole path at once. The result is usually broad, shallow, and not quite right. You end up rewriting most of it anyway.

Prompt chaining is the alternative. You break the work into multiple, sequential prompts. Each one builds on the output of the last. Rather than asking AI to do everything at once, you guide it through a structured workflow, checking the results at every stage.

> The power of a chain is not in any single link. It is in the pauses between them, where your judgement shapes what comes next.

If this sounds familiar, it should. Prompt chaining is the Iterate stage of the Conversation Loop (Chapter 6) made explicit. Each link in the chain is a moment where you review the output, apply your judgement, and steer what comes next. Without that review step, you are not chaining. You are just queuing up requests and hoping for the best.

10.1 Why It Works

Breaking a task into steps produces better results for the same reasons it works when humans do it.

Smaller steps mean more focused output at each stage, which improves reasoning. You review and adjust after each step, catching problems early rather than discovering them in the final product. You can change direction mid-process without starting over. Each step's output becomes the next step's input, so good foundations compound. And because you see the AI's reasoning at each stage, not just the final product, you stay in the conversation rather than outsourcing it.

10.2 Three Approaches

There is no single right way to chain prompts. The best approach depends on how much control you need and how predictable the process is. Here are three patterns, from least to most interactive.

10.2.1 Approach 1: Guided Workflow

When to use it: You want multi-step processing but prefer to stay in a single prompt.

How it works: Provide all steps upfront with explicit instructions. The AI processes them sequentially and returns structured output.

Example: Research methodology review

```
You are a research methodologist.
I'm writing a methods section for a paper
on employee retention in remote teams.

Follow these steps:
1. Identify the key variables I need to measure
2. Suggest appropriate data collection methods
3. Outline potential confounding variables
4. Recommend how to control for them

Here's my research question: [Paste research
question]

Output each step clearly labelled.
```

This works well when you have a clear process in mind and trust it enough to let the AI work through it in one pass. You still review the output. You just review it all at once rather than step by step.

10.2.2 Approach 2: Sequential Chain

When to use it: You want to review and adjust between steps. Each prompt depends on what the previous one produced.

How it works: Submit one prompt. Review the output. Then submit a follow-up that builds on it.

Example: Writing a proposal

Step 1, Analysis:

```
I'm writing a funding proposal
```

```
for a community health programme.
Here's a summary of the problem
we're trying to address:
[Paste summary]

Analyse this. What are the strongest
arguments for funding? What gaps
in the evidence should I be aware of?
```

Review the output. Note what resonates and what is missing.

Step 2, Structure:

```
Based on that analysis, outline a proposal
structure.
Include sections for: problem statement,
proposed approach, expected outcomes,
and evaluation plan.
For each section, note the key point it should
make.
```

Review. Adjust the structure before proceeding.

Step 3, Draft:

```
Now draft the problem statement
and proposed approach sections.
Use a direct, evidence-informed tone.
Keep it under 800 words total.
Emphasise the points we identified
in Step 1.
```

Review, revise, continue with remaining sections.

The value here is in the pauses. Each review point is a chance to correct course, add something the AI missed, or redirect entirely.

> **What the pause between steps looks like**
>
> Returning to the funding-proposal chain, here is the moment between Step 1 (Analysis) and Step 2 (Structure):

> **AI (Step 1 response):** Three strongest arguments for funding: (1) the catchment area shows measurable health-outcome gaps that match the programme's targets; (2) the delivery model has been piloted at smaller scale with documented uptake; (3) the partner organisations bring infrastructure that reduces the funder's operational risk. Two gaps in evidence: the pilot data is from a single site, and there is no formal cost-per-outcome figure yet.
>
> **You (pause, thinking):** The single-site pilot point is the one that will sink the proposal if Step 2 builds on the current framing. I cannot fix the data, but I can change how I structure around it.
>
> **You (Step 2 prompt):** Outline a proposal structure that addresses the three strongest arguments, but treat the single-site evidence as a known weakness up front, not a hidden risk. Put a short "Evidence base and its limits" section before the proposed approach. The structure should make the funder feel we have already done their scepticism for them.
>
> *Without the pause, Step 2 would have produced a generic four-section outline. With it, the chain absorbs an evidence weakness the human spotted and turns it into a structural feature of the proposal. The AI did not know which of its findings was the load-bearing one. The human did.*

10.2.3 Approach 3: Interactive Chain

When to use it: You want the AI to guide you through a process, pausing for your input at each stage.

How it works: You set up the process, then the AI works through it collaboratively, waiting for your feedback before continuing.

Example: Analysing survey data

```
You are a research analyst. We're analysing
```

```
survey data about workplace satisfaction
across three departments.

Here's how we'll proceed:
1. I'll share the survey summary
2. You identify the top 3 themes
3. I'll respond with feedback
4. You develop deeper analysis of the themes I
approve
5. We continue iteratively until complete

Here's the data:
[Paste survey summary]

Start with step 2. Then wait for my
response before continuing.
```

The AI responds with themes. You reply:

> "Theme 1 is spot on. Theme 2 needs more nuance. Can you break down the 'communication issues' into specific types? For theme 3, I'd like to explore the relationship between workload and satisfaction more deeply."

The AI refines and waits. You stay in the driver's seat.

This approach works especially well for analysis and research tasks where you cannot know in advance what the interesting findings will be. The chain emerges from the work itself.

10.3 Practical Templates

These are starting points, not scripts. Adapt them to your work. The structure matters more than the specific wording.

10.3.1 Analysis Chain

For working through data, documents, or any situation that needs structured examination.

```
Context: [What are we analysing?]
Goal: [What decision or insight do we need?]

Step 1: Summarise [the data, document, or
situation]
Step 2: Identify [key patterns, issues, or
opportunities]
Step 3: Assess [impact, risk, or significance]
Step 4: Recommend [specific actions or next steps]

Output each step clearly. After step 1, I'll
review before
we continue.
```

10.3.2 Argument Chain

For building a case, writing a position paper, or developing a systematic analysis.

```
Claim: [The main argument or recommendation]

Step 1: Define [key concepts or terms]
Step 2: Provide [evidence or examples]
Step 3: Address [counterarguments or limitations]
Step 4: Synthesise [into a clear position
statement]

Output each step with supporting detail.
I'll provide feedback after each step.
```

10.3.3 Planning Chain

For scoping a project, structuring a strategy, or working through any multi-stage plan.

```
Objective: [What are we trying to achieve?]
Constraints: [Timeline, budget, resources, scope
limits]
```

```
Step 1: Define [the key deliverables or
milestones]
Step 2: Identify [dependencies, risks, and
assumptions]
Step 3: Sequence [the work into phases or stages]
Step 4: Detail [the first phase
   with specific action items]

Output each step. I'll review and adjust
before we continue.
```

10.4 Advanced Techniques

Once you are comfortable with basic chaining, these techniques let you extract more from the same conversation.

10.4.1 Refinement at a checkpoint

Iteration is already part of the Conversation Loop. Chaining adds a dependency: decide what this step has established before passing its result onwards. Repeating "improve this" without checking the assumptions can propagate the same error through cleaner drafts.

10.4.2 Format Shifting

Take the same underlying content and chain it into different formats:

```
Prompt 1: "Analyse this project data."
   [Full analysis request]
[Get detailed analysis]

Prompt 2: "Now turn that analysis into:
- A 2-minute briefing for the executive team
- A one-paragraph update for stakeholders
- A set of action items for the project team"
```

The analysis does the work once; shifting the format makes it usable for three audiences.

10.4.3 Perspective Shifting

Rerun analysis from different viewpoints:

> "Now redo that analysis from the perspective of: a new customer encountering this product for the first time, a support team handling complaints, and a competitor evaluating the market. How does each perspective change the recommendations?"

It forces the AI to stress-test its own output, and it often surfaces blind spots you would not have found otherwise.

10.5 Managing Long Conversations

There is a practical reality that prompt chaining runs into: AI has a finite memory.

A model receives a bounded context, which the surrounding product may truncate or summarise. It can also fail to use information that is present. There is no universal number of exchanges after which a conversation becomes unreliable; length, content, model, and application behaviour all matter. Research on long-context tasks found sensitivity to where relevant information appeared. Liu et al., Lost in the Middle (https://arxiv.org/abs/2307.03172)

A few practical habits keep this under control:

Start new conversations for new topics. If your chain is done and you are moving to a different task, open a fresh session. Do not try to do everything in one conversation.

One task per prompt. When you ask the AI to do three things at once, it divides its attention across all three. The result is three shallow responses instead of one deep one. Break it up.

Summarise and hand off. When a conversation is getting long or key constraints become hard to track, ask the AI to summarise everything you have accomplished and decided so far. Copy that summary into a new conversation and continue from there. You reset the AI's attention without losing the thread.

Make context explicit every time. Do not assume the AI remembers what you said earlier. If a constraint matters for this step, state it again. "Remember, we are working within a six-week timeline and a team of four" takes five seconds to type and prevents the AI from drifting back to generic assumptions.

Watch for signs of drift. If the AI starts contradicting something it said earlier, or produces output that ignores a constraint you set, inspect the handover and sources; context management is one possible cause, not the only one. Summarise, hand off, and continue in a fresh session.

These habits are housekeeping, and they are what keep a chain holding together over ten steps rather than quietly falling apart at step six.

> The golden rule of chaining
>
> Review each step before moving to the next. Without that review, you are not chaining. You are queuing up requests and hoping for the best.

10.6 When Not to Chain

Not every task needs a chain. Skip it when:

- **The task is simple.** A single clear prompt is more efficient.
- **You are exploring, not building.** Free-form conversation sometimes beats structured chaining.
- **Context is thin.** Without enough input, chains do not add much value. Start with a better first prompt instead. RTCF (Chapter 9) helps here.

10.7 Best Practices

1. **Start clear.** Your first prompt sets the direction. Use RTCF framework to structure it.

2. **Review each step.** Do not blindly proceed. Check each output before moving on.
3. **Provide feedback.** Tell the AI what is working and what needs adjustment.
4. **Build incrementally.** Small steps beat big jumps.
5. **Save your chains.** A good chain is reusable. Document the ones that work.
6. **Know when to stop.** Iteration has diminishing returns. Stop when the acceptance criteria are met or further revision stops producing a useful change.

10.8 Independent practice: inspect the handover

Time: 15–20 minutes. Sketch a three-step workflow: select group sizes, prepare the timetable, draft the invitation. Use Chapter 13. For each step, write the input, output, and a check that must pass before the next step starts. Decide where an unconfirmed assumption must stop the chain.

Artefact: a three-step handover map. **Success:** check capacity before timing and both before communicating an invitation. No AI account is needed to complete the independent version.

Chapter 11

Eight Techniques for Deeper Thinking

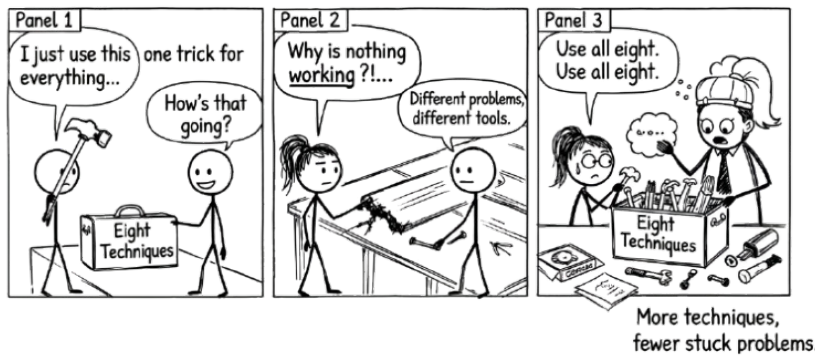


Figure 11.1: More techniques, fewer stuck problems.

> The right technique is the one that matches the kind of thinking you need to do.

AI works best as a thinking partner, not an answer machine, and your own thinking comes first. But what does that conversation actually look like?

This chapter gives you eight concrete techniques. Each one structures the conversation differently, pushing your thinking in a specific

direction. They cover most of the situations where people get stuck: scoping a problem, weighing options, reasoning through complexity, preparing for difficult interactions, stress-testing ideas, checking your own understanding, getting multiple perspectives, and finding blind spots.

For each technique, you will find a short explanation of what it does, when to reach for it, and a ready-to-use prompt you can adapt to your own situation.

11.1 1. Reverse Prompting

What it does. Instead of asking AI for answers, you ask AI to ask *you* questions. This forces you to articulate things you have not thought through yet. The AI's questions surface assumptions, gaps, and considerations you would otherwise miss. It works because a good question is often more valuable than a quick answer.

When to use it. You are starting a new project, scoping a problem, or planning something complex and you know you have not thought of everything yet.

A prompt to adapt:

```
I'm planning [describe your project,
decision, or problem].

Your task: Ask me a series of
short-answer questions to help me
clarify all the requirements,
considerations, and potential pitfalls.

Ask one question at a time. Wait for
my response before asking the next
question. Continue until you've helped
me think through at least 10 different
aspects of this.

Begin with your first question.
```

The key is to answer honestly, especially when a question catches you off guard. That discomfort is the technique working. If the AI asks something you cannot answer, that is a gap in your thinking you just discovered.

After the questioning round, try this follow-up: "Based on my answers, what is the single biggest gap in my thinking?"

Reverse Prompting works best at the start of something, when the shape of the problem is still forming. But once you have a clearer picture and need to choose a direction, you need a different kind of thinking.

11.2 2. Pros and Cons

What it does. The AI systematically evaluates multiple options against specific criteria you care about. The goal is a structured comparison that makes your own reasoning more rigorous. You bring the judgement. The AI brings the structure and breadth.

When to use it. You are choosing between approaches, tools, strategies, or solutions and you want to think through trade-offs methodically rather than going with your gut.

A prompt to adapt:

```
I'm deciding between these options
for [describe the decision]:

Option A: [describe]
Option B: [describe]
Option C: [describe]

For each option, provide:
1. A brief description of how it works
2. Three key advantages
3. Three key disadvantages

Evaluate each specifically in terms of:
```

```
- [Criterion 1, e.g., cost and time investment]
- [Criterion 2, e.g., long-term flexibility]
- [Criterion 3, e.g., risk if it goes wrong]

Conclude with which option you'd recommend
and why, but flag the strongest argument
against your recommendation.
```

Do not accept the AI's recommendation uncritically. The value is in the structured comparison. Challenge at least one of the listed pros. Ask yourself what disadvantage the AI missed. Argue for a different option than the one AI recommended and see if your reasoning holds up.

Pros and Cons gives you a snapshot of your options. But some decisions are not about choosing between options. They are about understanding a process with many moving parts, where the sequence matters and skipping a step has consequences.

11.3 3. Stepwise Chain of Thought

What it does. The AI walks you through a complex process one step at a time, pausing after each step. This prevents you from rushing through something that requires careful, sequential thinking. Each pause gives you a chance to ask questions, raise complications, or confirm you understand before moving on.

When to use it. You are learning a new process, preparing for something with many sequential steps, or working through a procedure where skipping a step could cause problems.

A prompt to adapt:

```
I need to understand the process
for [describe the task or procedure].

Walk me through the entire process
from start to finish. For each step,
tell me:
```

```
1. What action to take
2. What to document or record
3. What could go wrong at this stage

After you explain each step, STOP and
wait for me to type "next" before moving
to the next step. Do not provide
the entire process at once.

Begin with Step 1.
```

At any step, you can ask the AI to go deeper: "What if the client pushes back at this stage?" or "What happens if I skip this step?" These side conversations are where you stop memorizing a sequence and start understanding why each step matters.

The first three techniques help you think through problems, weigh options, and understand processes. But some of the hardest challenges are not analytical. They are interpersonal. You know what you want to say, but you are not sure how the other person will respond.

11.4 4. Role Play

What it does. The AI adopts a specific persona and you practice interacting with that persona. Think of it as a flight simulator for difficult conversations. The AI can play a skeptical stakeholder, a frustrated customer, a resistant colleague, or any other person you need to prepare for. It responds dynamically to what you say, so you get realistic practice without real consequences.

When to use it. You have a difficult conversation coming up, you want to practice a presentation to a tough audience, or you need to rehearse handling objections.

A prompt to adapt:

```
You are [describe the persona: e.g.,
```

```
a senior executive who is skeptical
of my proposal / a client who is
unhappy with a delayed deliverable /
a colleague who disagrees with my
approach].

Your personality: [e.g., direct and
impatient / friendly but unconvinced /
emotional and defensive]

I will practice having this
conversation with you. Stay in
character. If I say something vague,
push back. If I handle something well,
acknowledge it briefly but keep
pressing.

After we finish the conversation,
break character and give me feedback
on:
- What I handled well
- What I could improve
- Any phrases or approaches that
  would have been more effective

Let's begin. I'll speak first.
```

The feedback at the end is critical. It turns the exercise from mere practice into a structured debrief. Pay special attention to moments where the AI pushed back and you felt stuck. Those are the moments to rehearse differently.

Role Play prepares you for a specific conversation. But sometimes the challenge is a decision where you already have a preference and you suspect you have not fully examined the other side. That calls for a different kind of pressure.

11.5 5. Debating

What it does. You set up a structured debate where opposing positions are argued with equal force. This is powerful for decisions where you already have a preference, because it forces you to take the other side seriously. You can have AI argue both sides in a single conversation, or you can take one side yourself and let AI argue the other.

This technique combines well with using multiple AI tools. Give one model the "advocate" position and another the "skeptic" position, then shuttle arguments between them. Or keep it simple and have one AI play both roles.

When to use it. You are facing a decision with valid arguments on both sides and you suspect you are leaning one way without fully examining the alternative.

A prompt to adapt:

```
I'm considering [describe the decision].

Context:
- [Key fact 1]
- [Key fact 2]
- [Key constraint]

Set up a debate between two positions:
- "The Advocate" argues for [Option A]
- "The Skeptic" argues for [Option B]

Conduct the debate in four rounds.
Each speaker gets 3-4 sentences per
round. Label each speaker clearly.
Make both sides as persuasive as
possible.

After the debate, tell me: which
arguments should weigh most heavily
```

```
in my decision, and what additional
information would I need to be
confident either way?
```

Watch for arguments that surprise you. If the Skeptic raises a point you had not considered, sit with that. The purpose is not to "win" the debate, but to make sure you are not ignoring a perspective that matters.

A useful follow-up: "Now argue for a third option that synthesizes the best of both positions."

The previous five techniques help you explore, decide, learn processes, practise interactions, and stress-test positions. But sometimes the question is simpler and more personal: do I actually understand this topic as well as I think I do?

11.6 6. Formative Self-Testing

What it does. The AI generates questions that test your understanding of a topic, then gives you immediate feedback on your answers. The goal is to discover where your understanding is solid and where it is shakier than you thought. The act of retrieving knowledge and articulating it is one of the most effective ways to deepen your understanding.

When to use it. You are studying a new subject, preparing for a certification, onboarding into a new role, or just want to pressure-test how well you actually understand something you think you know.

A prompt to adapt:

```
I'm studying [topic or subject area].
Test my understanding.

Ask me 5 questions, one at a time. Mix the
difficulty:
- 2 questions testing core concepts
- 2 questions requiring application to realistic
```

```
scenarios
- 1 question that requires me to
  evaluate or critique something

After each answer I give, tell me:
1. Whether my answer is correct,
   partially correct, or incorrect
2. What I got right
3. What I missed or got wrong, with a brief
explanation

Wait for my answer before moving to the next
question.

Start with Question 1.
```

Be honest with your answers. Do not look things up first. The aim is to find out what you actually know versus what you think you know. Pay close attention to the "partially correct" answers. The gap between what you said and what you missed is where the most useful learning sits.

Self-testing reveals what you know and what you do not. But even when you understand a topic well, you are still seeing it from one vantage point: your own expertise, your own role, your own assumptions. Some problems need more than one perspective.

11.7 7. The Expert Panel

What it does. The AI simulates multiple experts with different backgrounds, each analysing your situation from their own perspective. A financial expert sees cost implications. A legal expert sees compliance risks. A customer experience expert sees user impact. You get multiple lenses on the same problem without needing to schedule five meetings.

When to use it. You are making a decision that affects multiple domains and you want to make sure you are not thinking about it

from only one angle.

A prompt to adapt:

```
I need to make a decision about [describe the
situation].

Assemble a panel of 4 experts who would
each have a different perspective
on this decision:
- Name each expert and describe
  their background
- Have each expert analyse the
  situation from their perspective
  (3-4 sentences each)
- Have each expert identify the one
  thing they think I'm most likely
  to overlook

After all four experts have spoken, provide a
synthesis:
what do they agree on, where do they disagree, and
what
should I investigate further?
```

The synthesis at the end matters most. Look for where the experts disagree. That tension usually points to the hardest part of your decision, the part that requires your judgement, not more AI output.

For a deeper conversation, pick one panelist and say: "I want to push back on [Expert Name]'s point about [specific claim]. Have them defend their position."

The Expert Panel broadens your perspective. But there is a particular kind of blind spot that multiple perspectives alone will not catch: the risks hiding inside your own confidence. When you have a plan you feel good about, you need someone to tell you what could go wrong.

11.8 8. Risk Deep-Dive

What it does. You share a plan or decision along with the risks you have already identified, and the AI helps you find what you are missing. It looks for second-order effects, blind spots, and early warning signs. This technique is specifically designed to counter the optimism bias that affects most planning. We tend to overestimate benefits and underestimate what can go wrong.

When to use it. You have a plan you are fairly confident about and you want someone to stress-test it before you commit.

A prompt to adapt:

```
I'm planning to [describe the initiative or
decision].

Here are the risks I've already identified:
- [Risk 1]
- [Risk 2]
- [Risk 3]

Help me think deeper:
1. What risks am I missing or underestimating?
2. What second-order effects should I consider?
3. Which risk is most likely to be the one that
actually
   derails this?
4. What early warning signs should I watch for?
5. What would have to be true for this to succeed?
   For each condition, say whether I have
   established it,
   merely made it plausible, or not proven it yet.
   Name
   the evidence that would move it up one
   category.

Challenge my assumptions. Be specific and
concrete.
```

The last question reverses the usual direction. Instead of asking only how the plan might fail, it makes success state its requirements. Keep the resulting list as an assumptions register rather than treating every plausible condition as a fact. An unproven condition is not automatically a reason to stop; it identifies what to check, who can check it, and whether the decision can proceed before it is known.

For critique, name the audience, deadline, reversibility, and stakes before reaching for a theatrical critic persona. A persona can help you rehearse a person (Technique 4), but "rip this apart" does not by itself create analytical rigour and can invite invented objections. The acceptance criteria do the serious work.

Question 1, the risks you are missing, usually produces the most useful output from this technique. We all have blind spots shaped by our experience, our role, and our optimism about our own plans. The AI does not share those blind spots.

After the initial analysis, try: "If this initiative fails 6 months from now, what is the most likely cause?" This "pre-mortem" framing often surfaces risks that a forward-looking analysis misses.

i What a pre-mortem actually surfaces

After running the Risk Deep-Dive on a proposed quarterly metric overhaul, the follow-up "If this initiative fails 6 months from now, what is the most likely cause?" landed this:

> **AI:** Most likely cause of failure in 6 months: the new metrics will technically launch on schedule, but middle managers will quietly continue reporting against the old metrics in their team reviews because the old ones map to bonuses that the metric change did not touch. By month four, leadership will be reading the new dashboard while the organisation is still being run on the old incentives. The failure will not be visible as a failure: it will look like adoption working but outcomes not moving.
> **You:** That is exactly what happened the last two

> times we changed metrics. We never touched the bonus tie-in because it sits with HR, not us. I need to talk to HR before launch, not after.
>
> *The forward-looking risk list had been about adoption, training, and dashboard tooling, all of it standard. The pre-mortem reframe surfaced the structural problem the forward list missed: the new metric and the existing incentive were owned by different teams. The fix is now a phone call, not a post-mortem.*

11.9 Combining Techniques

These eight techniques are not isolated tools. They combine naturally as a problem evolves.

You might start with **Reverse Prompting** to scope a new initiative, use **Pros and Cons** to evaluate your options, run a **Debate** to stress-test your preferred choice, convene an **Expert Panel** to check for blind spots across domains, and finish with a **Risk Deep-Dive** before committing.

Or you might use **Role Play** to prepare for a presentation, then **Formative Self-Testing** to make sure you can handle technical questions from the audience.

You do not need all eight on every problem, only the right one available when you need it.

> 💡 Start simple
>
> You do not need to master all eight techniques. Pick the one that fits your current problem. Try it once. Modify the prompt. Try it again. Skill comes from repetition, not from reading.

Start with whichever technique matches the kind of thinking you need to do right now.

11.10 Practice: choose, then justify

Time: 15–20 minutes. Choose one difficulty in a current task and select one technique from this chapter. Write why it fits and what would count as useful progress. Try it with a model, a colleague, or your own contrasting drafts, then record one change it produced.

Artefact: a technique choice and one before/after example. **Success:** explain why that technique was useful; do not use all eight merely to finish the checklist. No AI account is needed to complete the independent version.

Chapter 12

Using AI to Help You Use AI

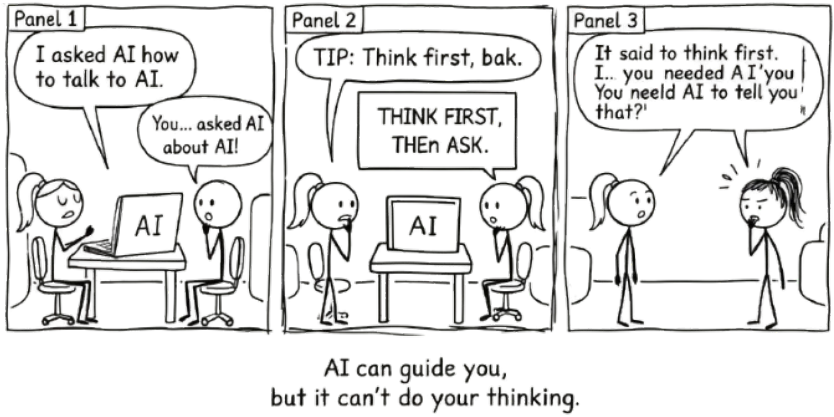


Figure 12.1: AI can guide you, but it can't do your thinking.

The most useful thing AI can do is help you figure out what you actually need to ask.

12.1 The One Thing AI Can Do That Other Tools Can't

AI can teach you how to use itself. Excel cannot teach you how to use Excel, PowerPoint cannot walk you through making better slides, and email cannot help you write better emails.

But AI can, because it can ask you questions, understand your context, evaluate its own suggestions, and adjust based on your feedback.

This means you do not need a course or a manual to get started. You can use AI to figure out how AI fits into your work.

12.2 The Core Idea: Meta-Prompting

Instead of asking AI to solve a problem directly, ask AI to help you figure out how to approach the problem.

Direct Approach	Meta Approach
"How should I answer this question?"	"What is the best way to frame this question for an AI so I get useful help?"
"Give me strategies for using AI."	"Ask me questions to understand my context, then recommend both obvious and unexpected ways I could use AI."
"How do I improve my workflow?"	"Interview me about my role and responsibilities, then suggest AI tools and strategies I might not have considered."

The meta approach works better because AI asks clarifying questions first, learns your context, and gives you recommendations that fit your actual situation rather than generic advice aimed at nobody in particular.

12.3 The Consultation Prompt: Your Starting Point

Here is a prompt you can use right now. Copy it, paste it into any AI tool, and start the conversation.

```
You are an AI expert consultant.
I would like your help understanding
how I could better use AI in my work.

Please ask me one question at a time.
I will answer, then you ask the next
question. Keep asking until you
understand:
- My role and responsibilities
- My main workflows and challenges
- My key objectives
  (what matters for success)
- My constraints (time, resources,
  skills, institutional requirements)

Once you have enough context,
provide TWO types of recommendations:
1. Obvious opportunities: Clear,
   straightforward AI applications
   (the low-hanging fruit most
   people think of)
2. Non-obvious opportunities:
   Unexpected or creative uses of AI
   I might not have considered

Format your recommendations clearly
with implementation tips for each.
```

12.4 Why This Works

When you ask AI to interview you before making recommendations, several useful things happen.

It gets your actual context. AI does not know anything about you at the start. By asking sequential questions, it builds up a picture of your discipline, your constraints, and what success looks like for you. Generic advice does not account for the fact that you manage compliance for a large team, or support early-career researchers, or run a small business with no IT budget. Context changes the recommendations entirely.

You get both obvious and non-obvious ideas. AI tends to suggest the predictable stuff first: automating reports, generating ideas, answering questions. Those are useful, but you have probably already thought of them. By explicitly asking for non-obvious recommendations, you push the model past its defaults. It starts connecting your specific constraints and workflows to capabilities you had not considered. The non-obvious ideas tend to be more useful, and they are the ones you would never have found by reading a generic "top ten AI tips" article.

AI can evaluate its own suggestions. Once it generates a list, you can ask it to assess which are most feasible, identify trade-offs, refine implementation steps, or suggest variations. This lets you iterate toward something practical rather than accepting the first answer.

It is a conversation, not a lecture. Unlike reading a guide or sitting through a workshop, you shape the advice as it forms. You answer honestly, push back on ideas that do not fit, and explore what is actually possible. The back-and-forth is where the useful stuff tends to surface. And because you are in control of the conversation, you can steer it toward the parts of your work that matter most.

12.5 A Real Example

Here is a simplified version of how this conversation might go.

You paste the consultation prompt.

AI asks: "What is your primary role, and what does a typical week look like?"

You say: "I am a teaching coordinator in a business faculty. I manage unit approvals, coordinate teaching support across 15 units, and help faculty integrate new teaching approaches."

AI asks: "What is your biggest pain point right now?"

You say: "Administrative overhead. Tracking who needs what support, writing emails summarising policies to faculty who have not read them, managing compliance documentation."

AI asks: "What would success look like for you?"

You say: "I would have time for strategic work instead of just admin. And faculty would feel confident designing assessments without constantly asking me the same questions."

AI synthesises:

> **Obvious opportunities:** Use AI to draft plain-language policy summaries. Use AI to help faculty write learning outcomes. Use AI to generate assessment design templates.
>
> **Non-obvious opportunities:** Use AI to analyse your incoming faculty questions and identify common gaps in understanding, then create targeted guidance before people ask. Use AI as a quality-check tool: paste a unit outline and ask it to flag compliance risks or unclear learning outcomes. Use AI to build a simple FAQ that handles the routine questions so you can focus on the ones that actually need your judgement.

The AI did not give generic suggestions about "leveraging AI for productivity." It understood the specific role, constraints, and frustrations, so the recommendations are grounded in something real. You could act on any of them this week.

12.6 What the Interview Makes You Say

Answering the questions forces you to articulate your own work.

By answering, you end up spelling out:

- What you actually do (versus what you think you do)
- What matters most (versus what is just urgent)
- What is possible (versus what feels impossible)
- What you have not tried (because you did not know it was an option)

The recommendations are often less useful than the explanation itself. Explaining your work to something that knows nothing about you makes you name the constraints, the trade-offs, and the parts you usually leave implicit.

People say this in different ways. "I did not realise how much time I was spending on that until the AI asked about it." "Just explaining my challenges helped me see a solution on my own." "The recommendations were fine, but the clarity about what I actually needed was the useful part."

The consultation prompt puts you in the position of expert and the AI in the position of interviewer. You know the answers. It asks the questions. Answering them often shows that you already knew what to do, and had not said it out loud.

12.7 Practical Tips for Your Consultation

Be honest and specific. Do not give polished answers. The more honest you are about your constraints and challenges, the better the recommendations. "We have no budget," "I do not have time to learn new tools," "My organisation is sceptical of AI." These are all useful context.

Ask follow-up questions. After getting recommendations, push further. "How would I actually implement the non-obvious ones?" "Which of these could I start with this week?" "What are the risks or drawbacks?"

Test one thing. Pick one obvious and one non-obvious recommendation. Try them for a week or two. Come back to the same conversation and report what happened. "I tried your suggestion about X. Here is what worked and what did not." The AI will adjust. Iterate from there.

Come back periodically. This is not a one-off exercise. Your context changes. Your needs shift. The tools themselves change. Run this consultation again in a few months. Each time, you will be ready for different ideas, and you will ask sharper questions because you have more experience to draw on.

Share what you learn. If something works, tell a colleague. "I asked AI to help me figure out how to use AI, and it suggested this. Here is how it went." Others might try their own version and find something entirely different that is useful for their context.

> **i** What the interview gives you
>
> The recommendations are often less useful than the explanation. Explaining your own work to something that knows nothing about you is what produces the clarity.

12.8 Getting Started Right Now

1. Open your AI tool.
2. Copy the consultation prompt from this chapter.
3. Paste it and start answering questions honestly.
4. Review the recommendations and pick one or two to try.
5. Come back with what you learned and keep going.

This works with any AI tool and in any role. The skill you are building is the ability to think clearly about your own work, explain it precisely, and evaluate whether what comes back is useful. That skill transfers everywhere.

If you want to structure the prompts that come out of this consultation, the RTCF framework (Chapter 9) gives you a reliable way to build them. If you want to break complex problems into stages,

prompt chaining (Chapter 10) shows you how. Once AI has helped you figure out what to ask, VET helps you make sure you can stand behind the answer.

12.9 Practice: evaluate the consultant

Time: 15–20 minutes. Write your own short brief before asking a model or colleague to improve it. Compare the revised brief against the original goal. Mark an assumption the consultant introduced, and retain it only if you can justify it.

Artefact: the two briefs with an acceptance/rejection note. **Success:** improve the request without letting the consultant quietly change what you are trying to achieve. No AI account is needed to complete the independent version.

Part IV

Part 4: Putting It Together

Chapter 13

Conversations Across Disciplines

A framework becomes useful when you can see the moment it changes a decision. This chapter follows one small community task from a first attempt through a bad AI proposal, a correction, an independent check, and a final plan. Every input is supplied. You can audit the exchange without opening an AI tool.

The scenario and dialogue are **constructed teaching examples**, not a transcript of a real event. The source cards below are authoritative within this fictional case; in a real project you would obtain the corresponding venue rules, quotations, and confirmations.

13.1 The task: a community seed workshop

Maya volunteers at a neighbourhood seed-sharing group. Eighteen people want a short workshop, and two volunteers will facilitate. She has a small budget and a ninety-minute room booking. She wants a plan she can explain to the other volunteer, not an attractive timetable with hidden assumptions.

Before asking AI, she writes:

> We need to serve 18 participants in 90 minutes. I think

> two groups could work. I have not checked whether the room limit includes us, and I need to preserve time to reset the tables.

That is her AI Last attempt: the goal, an initial approach, and the uncertainty. She is not required to solve the whole problem before asking for help.

13.1.1 Source cards

Card	Supplied fact
V1: venue booking	Room available 14:00–15:30; maximum 12 people in the room, including facilitators
V2: volunteer agreement	Two facilitators present for each session; minimum workshop duration 35 minutes; reset between groups takes 10 minutes
B1: budget	Maximum AUD 80; room costs AUD 30; shared supplies cost AUD 10; seed packs cost AUD 2 per participant
P1: attendance	18 participants; each attends one session; no participants enter early while another group is inside

The cards prevent the model from deciding what “capacity” or “budget” means. Maya can still change her approach, but she cannot change those facts merely because the timetable would be easier.

13.2 Brainstorm: make the request inspectable

Maya uses RTCF:

> **Role:** Help me plan a small community workshop.
>
> **Task:** Propose group sizes and a timetable that satisfy the four source cards. Show the capacity, time, and cost calculations.
>
> **Context:** There are 18 participants and two facilitators. Use V1, V2, B1, and P1 above. Do not invent a different room or extra money.
>
> **Format:** A short timetable, calculations, and any unresolved question. Do not draft the public invitation yet.

This separates planning from publishing. Asking for calculations does not ensure they will be right; it makes them easier to examine.

13.3 Ideate: a plausible proposal fails

The constructed first answer says:

> Run two groups of 12 and 6 participants, with both volunteers present. Each session lasts 45 minutes, with a 10-minute reset. This fits the 90-minute booking. Cost: $18 \times 2 + 30 + 10 =$ AUD 76.

One part is correct. The cost is 76. Two parts fail:

- Twelve participants plus two facilitators means **14 people**, above the limit of 12.
- Two 45-minute sessions plus a 10-minute reset means **100 minutes**, not 90.

Maya does not ask, "Are you sure?" She compares the proposed quantities with the cards. The budget answer's correctness does not transfer to the timetable or capacity answer.

Three levels of steering show the difference:

Response	What it changes
"Try again; make it better."	Signals dissatisfaction but leaves the actual errors unnamed
"Your timing is too long."	Identifies one failure, but leaves capacity unresolved
"V1 includes facilitators, so 12 participants makes 14 people. V2 makes your timetable 100 minutes. Recalculate both without changing the source cards."	Identifies the evidence, the errors, and the boundary for revision

The strongest intervention is specific enough that a different person could check it.

13.4 Iterate: repair one dependency at a time

Maya first asks for feasible group sizes. With two facilitators and a twelve-person limit, there can be at most ten participants at once. Two groups of nine fit: each session has eleven people in the room.

She checks that result before requesting a timetable. The next proposal is:

Period	Activity
14:00–14:35	Group 1: nine participants and two facilitators
14:35–14:45	Reset; first group leaves before the second enters
14:45–15:20	Group 2: nine participants and two facilitators
15:20–15:30	Pack-up/buffer within the booking

The chain makes the dependency visible: group size first, timing second, invitation last. If Maya lets the model draft the invitation before the checks, she risks polishing a plan that must be withdrawn.

A remaining practical question is how the second group will know when to arrive. P1 says no early entry, but does not provide an

accessible waiting area or a steward. Maya chooses to give the groups separate arrival times and asks the organiser to confirm the arrangement. She records that confirmation as outstanding rather than inventing it.

13.5 Verify, Explain, Test

Verify: Maya uses the cards and her own arithmetic, not the model's reassurance. Occupancy is $9 + 2 = 11$, within 12. Time is $35 + 10 + 35 + 10 = 90$ minutes. Cost is $18 \times 2 + 30 + 10 = 76$, leaving 4.

Explain: she tells the other volunteer why the first proposal failed, why facilitators count, and why reset time must appear in the schedule. She can explain the plan in speech, notes, or a sketch; fluent presentation is not the criterion.

Test: she changes one assumption. If one extra person arrives in each group, occupancy reaches 12 and the seed cost rises by 4, using the whole budget. The timetable still fits, but there is no spare budget. An additional arrival beyond that cannot simply be waved in. The plan needs a booking limit and an agreed response to walk-ins.

This is a small test of the plan's boundary, not a safety certification for a real event. A real organiser would also check relevant accessibility and venue requirements.

13.6 Amplify: own the final communication

Maya writes the final note:

> Two groups of nine, starting at 14:00 and 14:45. Each workshop lasts 35 minutes. Please arrive for your allocated session; the room limit includes the two facilitators. Planned cost is AUD 76 within the AUD 80 budget. Arrival arrangements still need the organiser's confirmation before we send invitations.

The note is less complete-looking than an invitation with invented assurances. It is more useful: another volunteer can see what is fixed and what is pending.

Her decision record keeps the initial attempt, the failed AI proposal, the source cards, the corrected calculations, the boundary test, and the outstanding confirmation. She does not need the entire chat transcript to show the important judgement.

13.7 Transfer to other kinds of work

A student can use the same sequence when a suggested argument claims more than its cited passage supports: name the claim, inspect the source, then decide whether to revise the argument.

For a project lead, the same sequence applies when an estimate assumes full-time staffing that is not available: correct the capacity before asking for a new delivery date.

It also helps a carer or community volunteer organise non-clinical schedules or communications while leaving health, safeguarding, and other specialist decisions with qualified people. The method helps expose dependencies; it does not supply professional authority.

In each setting, keep the distinction between generating an option, checking it, and authorising action. The conversation earns its place where it changes the work in a way you can explain.

13.8 Independent task: a changed workshop

Time: 25 minutes. No AI required. Use the same cards with these changes: **20 participants**, a **100-minute booking**, and a **minimum 40-minute session**. Keep the two facilitators, twelve-person room limit, ten-minute reset, and AUD 80 budget.

Prepare the plan before consulting the answer check. Show occupancy, timing, cost, and one boundary beyond which you would need to renegotiate a constraint. You may use a calculator, dictation,

large print, or a sketch. If the task feels large, complete capacity first and leave the other calculations for a second sitting.

Answer check: two groups of ten give 12 people per session. Two 40-minute sessions plus a 10-minute reset leave a 10-minute buffer. Cost is $20 \times 2 + 30 + 10 = 80$. There is no spare capacity or budget for another participant under that plan. Recognising a limit is part of the answer.

Success: all three calculations are visible, the original failed proposal is not reused, and the remaining practical confirmation is not invented. A polished timetable with an unexplained assumption does not pass.

13.9 Your Capstone: Produce an Owned Decision

Choose one real task whose outcome matters to you. It should be small enough to finish, but consequential enough that you would not accept a plausible answer blindly. Your deliverable is not the AI output. It is a one-page record of the judgement you exercised.

Begin without AI:

1. State the decision or outcome you need.
2. Write your starting position and what you already know.
3. Name what must be precise, what evidence would change your mind, and what you will not delegate.

Then run the task through the Conversation Loop. Keep only the exchanges that changed the work. Your record should show:

- your initial RTCF request;
- one response you rejected or materially redirected;
- the knowledge or value behind that intervention;
- one consequential claim you verified independently;
- the final decision in your own words; and
- what you can now explain or do without the AI that you could not before.

> **Capstone success check**
>
> The work is complete when another person can see where the AI contributed, where your judgement overruled or redirected it, what evidence you checked, and why you are prepared to own the result. A polished output without that trail does not pass.

If you cannot use an AI tool, run the same exercise with a colleague, a published argument, or two deliberately opposed drafts. The capability being assessed is critical conversation, not access to a particular product.

Chapter 14

When the Loop Runs Itself (Advanced and Optional)

> A useful test needs to check the thing you actually care about.

This optional chapter extends the conversation method to repeated, tool-using workflows. You can skip it and continue to Chapter 15 without missing the book's core practice.

A loop takes a goal, proposes an action, observes a result, and decides whether to try again or stop. Software can perform those repetitions quickly. What matters is how the goal, permissions, checks, and stopping conditions are specified.

14.1 Separate the loop from the authority

A workflow that drafts a reply and a workflow that sends it are different systems, even if their text is identical. A loop may be allowed to propose a change while a separate component decides whether the change is permitted.

Name every action available to the system. Reading a file can disclose information; writing one can damage a record; sending a message affects another person. "It only has tools" is not a small

claim. Each tool brings an access boundary and a possible consequence.

Also distinguish a fixed workflow from an agent that chooses its own next steps. Either may suit a task. More autonomy is not automatically more useful. Anthropic's practitioner account recommends starting with simpler patterns and adding complexity when justified by the work. Building effective agents (https://www.anthropic.com/engineering/building-effective-agents)

14.2 The test must sit outside the proposed answer

Suppose a loop edits a small program. An independently maintained test suite can check outputs for known inputs. That provides stronger evidence than asking the editing model whether its code is correct.

But a test suite is finite. Passing it does not establish every property of the program. A loop can exploit an incomplete metric, satisfy the examples while failing elsewhere, or alter the environment in a way the evaluator did not anticipate. Protecting the test file is helpful; it is not a universal safety proof.

For a writing task, a word-count limit is easy to check. Preservation of a contractual qualification is harder. If the loop learns to shorten by removing every caveat, it can improve its score while making the document less useful. You have optimised the measure, not necessarily the purpose.

14.3 A bounded loop specification

Before enabling an automated loop, record:

Element	Question to answer
Goal	What outcome should improve, and for whom?

Element	Question to answer
Inputs	Which sources and versions may be used?
Permissions	What may be read or changed, and what is prohibited?
Evaluation	Which independent checks must pass, and what do they miss?
Budget	Maximum attempts, time, cost, or resource use
Review gate	What requires a human decision before taking effect?
Stop and recovery	What halts the loop, who is notified, and how is the previous state restored?

A "keep going until done" instruction without these boundaries is not a plan. The system may be unable to reach the goal, unable to recognise completion, or rewarded for the wrong thing.

Use a disposable environment and synthetic inputs for practice. Do not test a stopping rule by allowing an agent to act on real customer accounts or messages.

14.4 Worked example: a shorter notice

Use this invented source notice:

> Group 2 starts at 14:45. Arrive for your allocated session. The room holds 12 people including facilitators. The organiser must confirm arrival arrangements before invitations are sent.

The loop's task is to shorten the notice while retaining the start time, capacity definition, and pending confirmation. A word-count check is one evaluator; a checklist of preserved facts is another.

A proposed revision says:

> Arrive at 14:45. The room holds 12 participants. Invitations are ready to send.

It is shorter. It also changes total occupancy into participant capacity and converts an outstanding confirmation into approval. A loop using only length would prefer it. A source-preservation check rejects it.

A better revision says:

> Group 2: 14:45. Room limit: 12 including facilitators. Send invitations only after the organiser confirms arrival arrangements.

A reviewer can trace each clause to the source. The approval gate still belongs outside the loop. Neither a good draft nor the model's statement "all checks passed" authorises sending.

14.5 Search can produce useful novelty

Repeated proposal and evaluation can discover combinations or solutions that were not explicitly supplied. There is no need to dismiss every such result as merely the appearance of creativity.

Equally, a novel-looking result does not establish validity. Ask what was new, how it was evaluated, whether the comparison was fair, and what assumptions the evaluation left untouched. Claims about what all future models could never discover are unnecessary to this decision.

The practical boundary is the quality of the evidence and the consequences of action. If the evaluator does not capture a requirement, repeated optimisation does not make that requirement disappear.

14.6 When the loop loses the thread

A long-running process needs records of completed work, current assumptions, failed attempts, and the remaining objective. That record may be stored outside a model's context. A summary can help, but it can also omit a constraint or preserve an error.

Check the handover at each restart. If a source changes, a tool fails, permissions expire, or the budget is reached, stop or escalate according to the specification. Do not let an unavailable check become an implicit pass.

A model-generated explanation of its behaviour can be useful evidence about what it says it did. Compare it with actual tool logs and outputs before relying on it. The prose is not an audit trail by itself.

14.7 Independent practice: inspect an optimiser

Time: 20 minutes. No model required. Use the notice above and imagine three checks: length under 30 words, correct time, and no sending without approval.

1. Mark what each check would catch in the bad revision.
2. Identify the missing capacity check.
3. Add a stop condition for a missing source or uncertain requirement.
4. Name which actions must remain unavailable during testing.
5. Write a candidate revision and check it against the original.

Artefact: a loop specification and checked notice. **Success:** preserve the capacity definition and pending approval, explain why length alone fails, and prohibit sending during the experiment. An automated extension may implement the checks, but the independent reasoning is the core task.

14.8 Where the human stands

Some bounded tasks can run with limited review; others need approval at a consequential step; some should not be automated in the available environment. Decide from the use, the evidence, and the failure cost.

That role may change as tools change. The book does not promise

permanent immunity from automation. It asks you to remain able to specify the task, inspect the evidence, notice a failed assumption, and account for the decision. A loop can accelerate parts of the work without making those questions unnecessary.

Chapter 15

Becoming More Capable

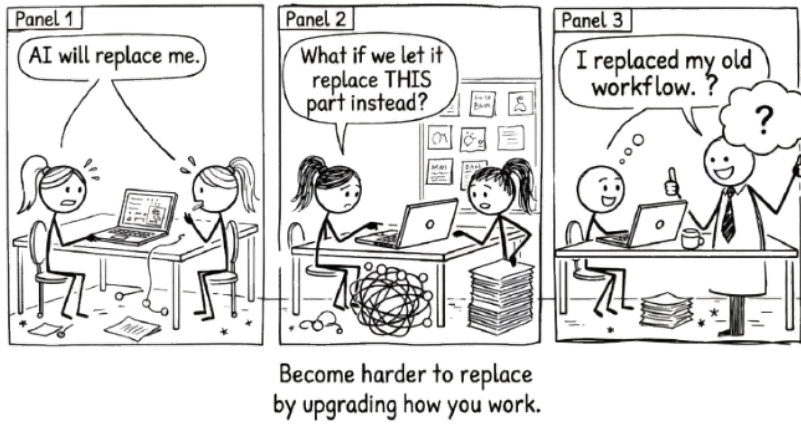


Figure 15.1: Become harder to replace by upgrading how you work.

> The goal was never to get AI to do more. It was to become more capable yourself.

15.1 Show the change, do not assume it

Return to the first attempt you kept near the beginning of this book. Put it beside the final version and explain what changed.

Perhaps you narrowed the question. Perhaps you rejected a confident assertion, checked a source, or noticed a limit that the proposed answer concealed. Those are observable actions. Finishing the chapters, holding a long conversation, or producing polished prose is not enough to establish them.

The evidence should be small enough to inspect: a starting position, an intervention, an independent check, and the decision you made. That is what the capstone asked you to produce.

15.2 The workshop, revisited

In the community-workshop case, the first proposal exceeded both the room capacity and the available time. The budget calculation happened to be right. Maya did not accept or reject the whole answer as a unit. She checked the parts against the source cards.

The corrected plan served eighteen participants in two groups of nine. It counted the facilitators, allowed reset time, and left the organiser's confirmation outstanding. The final message was useful because it preserved that limit.

The independent version changed the numbers. If you solved it without copying the worked timetable, you demonstrated transfer: the method survived a new case. If you missed a constraint, the check showed where to practise next. Both are more informative than whether an assistant congratulated you.

15.3 What remains yours

You choose the task and the standards that matter. You contribute knowledge of the setting, ask for missing evidence, decide what to accept, and identify who has authority to act.

AI can contribute candidate explanations, alternatives, transformations, and challenges. Sometimes it will supply an idea you would not have found alone. Sometimes a calculator, source document, colleague, or specialist will be the better resource. Choosing among

them is part of the capability this book is trying to build.

Do not interpret "own the result" as "I must perform every operation unaided." Good work often depends on tools and other people. Ownership means that the limits and responsibilities are visible and that your claim does not exceed what you can support.

15.4 A review of your own practice

Choose one piece of work from the last month. Ask four questions:

1. What did I believe before receiving assistance?
2. Which suggestion did I accept, reject, or materially change, and why?
3. Which consequential claim did I check somewhere other than the answer itself?
4. What can I now explain or do on a new example?

If you cannot answer one, identify a next action rather than writing a more flattering account of the process. You may need a source check, a simpler exercise, more domain knowledge, or another person's review.

Keep uncertainty in the record. "The source supports the schedule, but the organiser has not confirmed access" is stronger than pretending everything is settled.

15.5 A sustainable next habit

Choose one habit for the coming week, not a promise to use every technique on every task.

You might begin important work with a five-minute first attempt, keep a claim/source column beside a draft, place a deliberate pause between planning and execution, or explain a result in your own words before sharing it.

Choose a task where the habit has a visible completion test. At the end of the week, examine one example. Did the pause expose a real

issue? Did the source support the claim? Did you understand the change you accepted? Adjust the habit using that evidence.

There is no guarantee that each exchange compounds into lasting expertise. That is why the learning chapter separated assisted performance from independent transfer. Continue practising the capabilities you want to retain and check them on work that matters.

15.6 Capability is not an employment forecast

These habits do not establish what employers will value, which roles will change, or what future models will do. Promises that human judgement is permanently unautomatable are as difficult to defend as promises that every role will disappear.

The narrower reason to practise is already sufficient: today you need to make decisions under uncertainty, communicate clearly, and avoid giving unsupported outputs more authority than they deserve. Those are useful aims without an economic guarantee attached.

If a task is routine, well specified, permitted, and independently checkable, sensible delegation may be appropriate. If it needs discussion, converse. If the evidence or permission is missing, stop or use another method. The decision map in the quick reference keeps all of those options available.

15.7 Take stock without AI

Time: 20 minutes. Close the chat and choose a new, small task. Write the goal, relevant constraints, an initial approach, and how you would check it. You may use notes, a calculator, assistive tools, or an accessible format.

Compare your attempt with the record you made at the start of the book. Identify one improvement with evidence and one capability that still needs practice. Ask a colleague to challenge your explanation if someone is available; otherwise perform a second review after a break.

Success: you can show a specific change in how you approached or checked the work. You do not need to claim that the book transformed your intelligence or made you irreplaceable.

The next useful conversation begins with that honest account of where you are.

Appendix A

Prompt Structuring Frameworks

RTCF is not the only prompt structuring framework. A family of mnemonics emerged in 2023-24 as practitioners and educators figured out what makes a good opening prompt. They all capture the same insight: telling the AI who to be, what to do, what background it needs, and how to format the output produces better results than an unstructured request.

This appendix compares the most widely used frameworks side by side, using the same task so you can see how each one shapes the prompt differently. The task: getting AI to help analyse student feedback data.

None of these frameworks have a single academic source. They are practitioner knowledge, patterns that emerged from use, were shared in workshops and online communities, and proved useful enough to stick. Use them as reminders to supply relevant information, not as a guarantee that a longer or more structured prompt performs better. The mnemonics are packaging, not discovery.

We chose RTCF for this book because it has the fewest elements to remember and covers the ground that matters most. But if a different framework clicks better for how you think, use that one. What matters is the habit of thinking before prompting.

A.1 RTCF: Before and After

To see why structure matters, compare these two prompts for the same task.

⚠ Weak prompt

"What can you tell me about student feedback?"

RTCF prompt

"You are an experienced higher education analyst **(R)**. Identify the three strongest themes and two areas of concern in the student feedback data I will paste below **(T)**. This is from a second-year undergraduate unit with 85 responses across two campuses **(C)**. Present findings as a summary paragraph followed by a table with columns: Theme, Evidence, and Recommended Action **(F)**."

You do not need all four elements every time; use what is relevant. But when a prompt feels vague and you are not sure why, checking against RTCF may help identify missing information.

A.2 The Family

A.2.1 RTCF: Role, Task, Context, Format

The framework used throughout this book. Four elements, no extras. Role sets the expertise. Task specifies the action. Context provides the background. Format defines the output structure.

Same task:

> "You are an experienced higher education analyst **(R)**. Identify the three strongest themes and two areas of concern in this student feedback data **(T)**. This is from a second-year undergraduate unit with 85 responses across two campuses **(C)**. Present as a summary paragraph followed by a themed table **(F)**."

A.2.2 CRAFT: Context, Role, Action, Format, Tone/Target

Adds a fifth element: Tone or Target audience. Useful when the same content needs to be pitched differently for different readers, such as a board summary versus a team debrief.

Same task:

> "This is student feedback data from a second-year undergraduate unit with 85 responses across two campuses **(C)**. You are an experienced higher education analyst **(R)**. Identify the three strongest themes and two areas of concern **(A)**. Present as a summary paragraph followed by a themed table **(F)**. Use clear, non-technical language suitable for a faculty teaching committee **(T)**."

A.2.3 CO-STAR: Context, Objective, Style, Tone, Audience, Response

Six elements. Splits what RTCF bundles into Context and Format into finer-grained components. Useful for communication-heavy tasks where style, tone, and audience are genuinely distinct decisions.

Same task:

> "I have student feedback data from a second-year undergraduate unit, 85 responses across two campuses **(C)**. Identify the three strongest themes and two areas of concern **(O)**. Write in the style of an institutional quality report **(S)**. Professional but accessible, avoiding jargon **(T)**. For a faculty teaching committee that includes both academics and professional staff **(A)**. Summary paragraph followed by a themed table with columns: Theme, Evidence, Recommended Action **(R)**."

A.2.4 RISEN: Role, Instructions, Steps, End goal, Narrowing

Five elements with a distinctive feature: it asks you to specify the steps the AI should follow and the end goal you are working toward. Useful for multi-stage tasks where the process matters as much as the output.

Same task:

> "You are an experienced higher education analyst **(R)**. Analyse the student feedback data I will paste below **(I)**. First, read all responses and identify recurring themes. Then, categorise each theme as a strength or concern. Finally, assess which findings are actionable **(S)**. The end goal is a briefing document for a faculty teaching review **(E)**. Focus only on themes that appear in at least 10% of responses **(N)**."

A.2.5 CREATE: Character, Request, Examples, Adjustment, Type, Extras

Six elements with a distinctive feature: it builds iteration directly into the framework. The Adjustment step acknowledges that the first response is rarely the final one, and the Examples step lets you show the AI what you want rather than just describing it.

Same task:

> "Act as an experienced higher education analyst **(Character)**. Analyse this student feedback data and identify the strongest themes and areas of concern **(Request)**. Here is an example of the format I want: 'Theme: Student engagement | Evidence: 15 responses mentioned limited interaction | Action: Introduce weekly discussion activities' **(Examples)**. The initial output should focus on the three most prominent themes; I will refine the scope after reviewing **(Adjustment)**. Present as a themed table **(Type)**. Include a brief note on statistical confidence where relevant **(Extras)**."

A.2.6 APE: Action, Purpose, Expectation

Three elements. The most minimal framework. Useful for quick, low-stakes prompts where brevity matters more than precision.

Same task:

> "Analyse this student feedback data **(A)** to help me prepare for a teaching review meeting **(P)**. I expect a short summary of key themes and a table of findings with recommended actions **(E)**."

A.3 Choosing a Framework

Table A.1: Prompt structuring frameworks compared. More elements give finer control but take longer to compose.

Framework	Elements	Best for
APE	3	Quick tasks where speed matters more than precision
RTCF	4	General-purpose prompting that covers most situations
CRAFT	5	Communication tasks where tone and audience are distinct decisions
RISEN	5	Multi-step tasks where the process matters
CREATE	6	Tasks where you can show examples and expect to iterate

Framework	Elements	Best for
CO-STAR	6	High-stakes communication with specific style and audience requirements

The frameworks are listed from simplest to most detailed. Start with RTCF. If you find yourself repeatedly needing to specify tone separately from context, try CRAFT. If you need to define a multi-step process, try RISEN. If you have a clear example of what good output looks like, try CREATE. If you never need more than three elements, APE is fine.

The goal is to build the habit of structuring your thinking before you type. Any of these frameworks will get you there. Pick one, use it until it becomes instinct, then let it go.

A.4 The Caveat

Every one of these frameworks improves your *opening* prompt. None of them replace the conversation that follows. A perfectly structured RTCF or CO-STAR prompt is still a single prompt. It is still one shot. The value comes from what you do with the response: iterate, push back, refine, and make the output yours.

Appendix B

Quick Reference Cards

Tear-out summaries of the frameworks, tools, and techniques in this book. Pin them next to your screen or keep them open in a tab.

B.1 RTCF Prompt Framework

```
                RTCF PROMPT FRAMEWORK


  R - ROLE       Who should the AI be?

                 "You are a [expert type]..."


  T - TASK       What should it do?

                 Use action verbs: Analyse,

                 Compare, Create, Evaluate...


  C - CONTEXT    What does it need to know?
```

```
                Industry, constraints, audience

F - FORMAT      How should output look?

                Structure, length, style
```

Four components. Four questions. Answer them before you type your prompt, and the conversation starts in a much better place. See Chapter 9 for the full chapter.

B.2 VET Your AI

Three steps to run before you trust any AI output that matters.

Table B.1: The VET framework: Verify, Explain, Test.

Step	Question	What you do
V, Verify	Can I find this independently?	Cross-reference claims against credible sources. If the AI cites something, look it up.
E, Explain	Can I explain this in my own words?	If you cannot walk someone through the reasoning, you do not understand it well enough to use it.
T, Test	Does this hold up under scrutiny?	Challenge the output. Change the assumptions. Ask "what if the opposite were true?"

VET in practice:

Table B.2: VET applied to real AI output. The domain does not matter. The habit does.

AI says…	V: Verify	E: Explain	T: Test
"Studies show pH drops faster in co-culture"	Find the actual studies	Can I explain the mechanism?	What about different substrates?
"Optimal temperature is 37C"	Check the literature	Why 37C specifically?	What happens at 35C or 40C?
"This method has 95% recovery rate"	Where is that number from?	What does 95% mean here?	Under what conditions?

If you pass all three steps, the output is yours. If you fail any of them, go back into the conversation and iterate. See Chapter 8 for the full chapter.

B.3 Objective Review of a Proposal

Use this compact sequence when a fresh evaluation matters more than continuing the conversation that produced the idea. It combines the controlled handover in Section 6.7.1, VET in Chapter 8, and the risk work in Chapter 11.

Table B.3: An objective review need not use every step. Choose the shortest sequence that exposes the decision's weak point.

Step	Do this	Keep
1. Consolidate	From the exploratory work, write a short brief: decision, audience, deadline or reversibility, supplied evidence, constraints, unknowns	Brief checked against original sources
2. Reset context	Start a fresh approved session and supply only the brief	No exploratory turns or discarded framing
3. Surface omissions	Ask diagnostic questions one at a time about material left out	Answers, including "unknown"
4. Stress-test	Ask for the strongest specific objections and what evidence would change them	Objections tied to claims, not theatrical criticism
5. Name success conditions	Ask what would have to be true; mark each condition established, plausible, or unproven	Assumptions register with an owner and next check

Step	Do this	Keep
6. Compare, then revise	Optionally give the identical brief to a second approved model or qualified reviewer; investigate recurring and divergent risks	Comparison notes; overlap treated as a lead, not independent proof
7. Record movement	When you or the reviewer changes position, name the new evidence that caused it	Revision ledger: original claim, pushback, evidence, revised claim

A prompt to adapt

"Above is a self-contained brief for a decision. Write for the named audience and take its deadline and reversibility seriously. First ask me one material question at a time about information the brief omits. Then make the strongest specific case against the proposal, distinguishing supplied evidence from hypotheses. Finally, list what would have to be true for it to succeed and mark each condition established, plausible, or unproven, with the evidence that would move it up one category. If a later claim changes your assessment, name that claim; I will record the change in my own revision ledger and verify it independently."

A fresh session is not a zero-bias room: the brief can carry the original framing with it. Check current product guidance for any temporary/private/incognito label before relying on what it disables, and never move unapproved data into a second service merely to obtain another opinion.

B.4 The Conversation Loop

Four stages. One feedback arc. Most good work passes through more than once.

Table B.4: The Conversation Loop: Brainstorm, Ideate, Iterate, Amplify.

Stage	What happens	Your mindset
Brainstorm: Define your question	Arrive with a question, not a task.	"What am I actually trying to figure out?"
Ideate: Explore possibilities	Go wide with the AI.	"What angles haven't I considered?"
Iterate: Push back and refine	Challenge, redirect, sharpen.	"That's close, but here's what's wrong…"
Amplify: Make it yours	Fold in what you know. Own the result.	"How do I bring in what the AI doesn't know?"

The loop is not linear. You can jump from Iterate back to Brainstorm when you realise the question was wrong. You can loop from Amplify back to Ideate when a new angle emerges. The sign that you are done is not that the AI stopped producing output. It is that your thinking has landed somewhere solid. See Chapter 6 for the full chapter.

B.5 Eight Techniques at a Glance

Table B.5: Eight techniques for deeper thinking with AI.

#	Technique	What it does in one line
1	**Reverse Prompting**	The AI asks *you* questions to scope a problem before you solve it.
2	**Pros and Cons**	Systematic evaluation of competing options against specific criteria.
3	**Stepwise Chain of Thought**	Forces the AI to show its reasoning one step at a time so you can check each link.
4	**Role Play**	The AI adopts a stakeholder perspective so you can rehearse conversations and anticipate objections.
5	**Debating**	The AI argues the opposite position, strengthening your reasoning through challenge.
6	**Formative Self-Testing**	The AI generates practice questions and gives immediate feedback on your answers.
7	**Expert Panel**	Multiple simulated experts weigh in from different disciplines, forcing you to synthesise.
8	**Risk Deep-Dive**	Structured identification and analysis of risks, blind spots, and failure modes.

Each technique is a different way to stay in conversation rather than delegating. Pick the one that matches your task. See Chapter 11 for detailed walkthroughs and examples.

B.6 One Idea, Several Names

The same principle appears across disciplines under different vocabularies. Whichever language fits your audience, the practice is the same.

Table B.6: One idea, four vocabularies.

Name	Where you will hear it	What it means
Conversation	This book	Stay engaged at every stage; push back, iterate, own the output.
Judgement	Professional practice, education	The capacity to weigh, decide, and stand behind a conclusion.
Agency	Philosophy, ethics, policy	The capacity to choose, act, and be accountable for the outcome.
Human-in-the-loop (HIL)	AI governance, system design	A person remains an active participant in decisions a system could otherwise automate.

Conversation is the practice. Judgement is what you exercise during it. Agency is what you preserve by exercising judgement. Human-in-the-loop is what the same stance looks like from a system designer's perspective.

B.7 Choose the mode before the prompt

Situation	Starting mode	Required check
Routine calculation or exact transformation with a known method	Use a calculator, script, or established non-AI tool	Validate inputs and result

Situation	Starting mode	Required check
Well-specified, permitted task with an independent acceptance test	Delegate a bounded operation	Keep permissions narrow, inspect failures, preserve a stop rule
Ambiguous goal, competing values, or a judgement you need to develop	Converse	Preserve alternatives, evidence, and your decision
Repeated task with protected checks and recoverable actions	Consider automation	Test the evaluator, budget, access boundaries, and rollback
Data use is unapproved, evidence is missing, or consequences exceed your expertise	Avoid that AI use or seek qualified help	Obtain permission or expertise before proceeding

These are starting decisions, not permanent labels. Reconsider when the data, use, or consequences change.

Appendix C

Further Reading

This is a selective evidence trail, checked for this revision on 14 September 2026. Research findings below concern the settings studied; they are not guarantees about every learner or future model. Product documentation describes a changing implementation. Follow the direct links to inspect methods and limitations.

C.1 How models generate and use information

- **Vaswani et al. (2017), Attention Is All You Need (https://arxiv.org/abs/1706.03762).** Introduces the transformer architecture. Useful background for attention and causal decoding, not evidence that a model cannot revisit earlier prompt tokens or that all present-day systems share one architecture.
- **Lewis et al. (2020), Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (https://arxiv.org/abs/2005.11401).** Describes a system combining a learned generator with retrieval. Supports distinguishing model parameters from retrieved context; it does not establish that retrieval eliminates false answers.
- **Carlini et al. (2021), Extracting Training Data from Large Language Models (https://arxiv.org/abs/2012.07805).** Demonstrates recoverable training examples in the studied

setting. A reason to avoid claiming that generation can never reproduce stored material, not a claim that every response is a retrieved copy.

- **Liu et al. (2023), Lost in the Middle (https://arxiv.org/abs/2307.03172).** Tests how the position of relevant material affects performance in long contexts. Relevant to handover briefs and source placement; a large context window alone does not establish effective use of every passage.

C.2 Learning and critical thinking

- **Lee et al. (2025), The Impact of Generative AI on Critical Thinking (https://www.microsoft.com/en-us/research/wp-content/uploads/2025/01/lee_2025_ai_critical_thinking_survey.pdf).** Survey evidence about knowledge workers' reported effort and confidence. Self-report and association do not establish that AI caused a loss of general reasoning ability.
- **Bastani et al. (2025), Generative AI without guardrails can harm learning (https://doi.org/10.1073/pnas.2422633122).** Studies AI assistance in high-school mathematics, including performance when help is removed. The design of assistance matters; transfer to other subjects and workplace tasks requires further evidence.

These sources motivate separating task performance, learning, and later independent performance in Chapter 4. They do not prove that every copied answer harms learning, that every conversation helps, or that accessibility support should be withdrawn. If the PNAS publisher is inaccessible, the authors also provide an open manuscript (https://hamsabastani.github.io/education_llm.pdf); check the version against the published record.

C.3 Checking agreement and protecting information

- **Sharma et al. (2023), Towards Understanding Sycophancy in Language Models (https://arxiv.org/abs/2310.13548).** Studies agreement with user beliefs and preference-training effects. Treat sycophancy as a behaviour to test, not an unchangeable property of all assistants.
- **Cheng et al. (2025), ELEPHANT: Measuring and Understanding Social Sycophancy in LLMs (https://arxiv.org/abs/2505.13995).** Measures face-preserving responses in open-ended advice across eleven models, including affirmation of either side in paired moral-conflict prompts. Useful for distinguishing social affirmation from factual agreement; its rates describe the benchmark's datasets and measures, not every workplace conversation.
- **OAIC, Guidance on privacy and commercially available AI products (https://www.oaic.gov.au/privacy/privacy-guidance-for-organisations-and-government-agencies/guidance-on-privacy-and-the-use-of-commercially-available-ai-products).** Australian regulatory guidance for assessing personal information and AI use. It is not a product certification or advice about your particular obligations.
- **OAIC, De-identification and the Privacy Act (https://www.oaic.gov.au/privacy/privacy-guidance-for-organisations-and-government-agencies/handling-personal-information/de-identification-and-the-privacy-act).** Explains why identifiability depends on context. Replacing a name is not sufficient evidence that a record is anonymous.

C.4 Prompting, refinement, and workflows

- **Wei et al. (2022), Chain-of-Thought Prompting Elicits Reasoning in Large Language Models**

(https://arxiv.org/abs/2201.11903). Reports improvements with worked reasoning examples on the tasks and models tested. It does not show that generated explanations faithfully reveal internal reasoning.

- **Zhou et al. (2023), Least-to-Most Prompting Enables Complex Reasoning in Large Language Models (https://arxiv.org/abs/2205.10625).** Tests decomposition into ordered subproblems. Relevant to chaining, while leaving open whether a particular real-world task benefits from more stages.
- **Madaan et al. (2023), Self-Refine: Iterative Refinement with Self-Feedback (https://arxiv.org/abs/2303.17651).** Evaluates an iterative feedback method on several tasks. Improvements in those evaluations do not guarantee that repeated self-critique finds factual mistakes.
- **Anthropic, Building Effective Agents (https://www.anthropic.com/engineering/building-effective-agents).** Practitioner engineering guidance on workflows and agents. Use it to inspect permissions, tool execution, and evaluation, not as independent evidence of a universal productivity gain.

C.5 What belongs to this book

RTCF (**Role, Task, Context, Format**), VET (**Verify, Explain, Test**), AI Last, and the conversation loop are practical organising frameworks in this book. They combine familiar briefing, checking, and reflective practices. This edition does not claim a unique historical origin, a clinical learning intervention, or universal experimental validation for any of them.

The workshop in Chapter 13 and the short exchanges elsewhere are constructed teaching cases unless explicitly identified otherwise. Their arithmetic can be checked from the printed inputs; their dialogues are not transcripts or measured model performance.

A useful next reading step is to choose one claim that matters to your own work, open its source, and record what the study actually measured and one thing it leaves unresolved.

About the Author



Figure C.1: Photo of Michael Borck

Michael Borck is a software developer and educator working at the intersection of human expertise and artificial intelligence. He developed the Conversation, Not Delegation methodology: the idea that AI is most valuable not as a tool you delegate to, but as a thinking partner you converse with.

The methodology grew out of a realisation: focusing on crafting the perfect prompt was just another form of delegation. The real value was never in the prompt. It was in the conversation that followed: the pushback, the iteration, the moments where your own thinking sharpened because you stayed engaged. That insight shifted the focus from prompt engineering to something closer to how professionals have always done their best work: through structured dialogue, critical evaluation, and judgement that only comes from experience.

Michael applies these principles across software development, education, and creative projects. He creates educational software and resources, and explores the 80/20 principle in learning and productivity.

Connect

- GitHub projects (https://github.com/michael-borck): software and educational resources
- Free book collection (https://books.borck.education): current online editions and downloads
- 8020workshop.com (https://8020workshop.com): passion projects and workshops
- LinkedIn (https://linkedin.com/in/michaelborck)

Other Books in This Series

Foundational Methodology:

- *Converse Python, Partner AI: The Python Edition*

Python Track:

- *Think Python, Direct AI: Computational Thinking for Beginners*
- *Code Python, Consult AI: Python Fundamentals for the AI Era*
- *Ship Python, Orchestrate AI: Professional Python in the AI Era*

Web Track:

- *Build Web, Guide AI: Business Web Development with AI*

For Educators:

- *Partner, Don't Police: AI in the Business Classroom*